



Degree Project in Applied Mathematics and Industrial Engineering and Management

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Dynamic Optimization of Individual Financial Well-Being

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Abstract

This thesis develops a dynamic programming framework for individualized financial advisory in a Swedish banking context. The study focuses on young adults aged 18–30 who face several competing financial goals, including maintaining an emergency buffer, saving for a housing down payment, building short-term savings, and accumulating long-term savings. The purpose is to examine how financial well-being can be represented through measurable financial goals and used in a dynamic optimization model. The aim is to study whether the model produces financially interpretable allocation behavior under exogenous risks.

Financial well-being is represented through measurable components. These components are aggregated into alternative utility specifications and embedded in a finite-horizon dynamic programming model. The model incorporates stochastic elements such as unemployment, financial crises, income variation, and short-term spending. Parameter values are calibrated using Swedish public statistics, official institutional sources, return assumptions from Handelsbanken, and relevant literature. The model is solved numerically in Python, and the resulting policy allocations are evaluated using Monte Carlo simulations and sensitivity analysis.

The results suggest that the model can represent trade-offs between liquidity, goal attainment, and long-term financial development. Specifications with a stronger weight on housing savings reach the down payment goal more frequently and faster, while more balanced specifications perform better when income is constrained and financial resilience becomes more important. The inclusion of a housing-goal bonus further strengthens goal-oriented behavior and improves down-payment outcomes. However, the results are sensitive to parameter choices, particularly the utility weights and the housing bonus.

The thesis illustrates how dynamic programming can provide a structured and transparent framework for analyzing financial allocation decisions across several competing goals. While the model relies on simplifying assumptions and a limited definition of financial well-being, it illustrates how optimization-based methods could serve as a conceptual basis for more individualized advisory tools in banking practice.

Keywords

Dynamic programming, Financial well-being, Household finance

Sammanfattning

Denna uppsats utvecklar ett dynamiskt programmeringsramverk för individanpassad finansiell rådgivning i en svensk bankkontext. Studien fokuserar på unga vuxna i åldern 18–30 år, som står inför flera konkurrerande finansiella mål däribland att upprätthålla en buffert, spara till en kontantinsats för bostad, bygga upp kortsiktigt sparande och ackumulera långsiktigt sparande. Syftet är att undersöka hur finansiellt välbefinnande kan representeras genom mätbara finansiella mål och användas i en dynamisk optimeringsmodell. Målet är att studera om modellen genererar finansiellt tolkningsbara allokeringsbeslut under exogena risker.

Finansiellt välbefinnande representeras genom mätbara komponenter. Dessa komponenter aggregeras i alternativa nytto-specifikationer och integreras i en dynamisk programmeringsmodell med ändlig tidshorisont. Modellen inkluderar stokastiska inslag såsom arbetslöshet, finansiella kriser, inkomstvariation och kortsiktigt spenderande. Parametervärden kalibreras med hjälp av svensk offentlig statistik, officiella institutionella källor, avkastningsantaganden från Handelsbanken och relevant litteratur. Modellen löses numeriskt i Python och de resulterande allokeringarna utvärderas med hjälp av Monte Carlo-simuleringar och känslighetsanalys.

Resultaten visar att modellen fångar meningsfulla avvägningar mellan likviditet, måluppfyllelse och långsiktig finansiell utveckling. Specifikationer med en starkare vikt på bostadssparande når kontantinsatsmålet oftare och snabbare medan mer balanserade specifikationer presterar bättre när inkomsten är begränsad och finansiell motståndskraft blir viktigare. Inkluderingen av en målbonus för uppnådd kontantinsats förstärker ytterligare målinriktat beteende och förbättrar utfallet för målet. Resultaten är dock känsliga för parameterval, särskilt nytto-vikterna och bostadsbonusen.

Uppsatsen visar att dynamisk programmering kan utgöra ett strukturerat och transparent ramverk för att analysera finansiella allokeringsbeslut mellan flera konkurrerande mål. Även om modellen bygger på förenklande antaganden och en avgränsad definition av finansiellt välbefinnande, illustrerar den hur optimeringsbaserade metoder skulle kunna stödja mer individanpassade rådgivningsverktyg i bankverksamhet.

Nyckelord

Dynamisk programmering, Finansiellt välbefinnande, Hushållsekonomi

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Contents

1	Introduction	1
1.1	Background	1
1.2	Problem	2
1.3	Purpose	3
1.4	Research Question	3
1.5	Research Methodology	4
1.6	Disposition	4
2	Background	5
2.1	Dynamic Programming	5
2.2	Monte Carlo Simulation	6
2.3	Literature Review	7
2.3.1	Financial advice for young adults	7
2.3.2	Financial well-being measurement	8
2.3.3	Goal-based wealth management	8
2.3.4	Financial crisis as an exogenous risk	9
2.3.5	Precautionary saving and buffer-stock models	10
2.4	Summary	10
3	Methodology	12
3.1	Research Process	12
3.1.1	Definition of financial well-being	12
3.1.2	Selection of financial dimensions	13
3.1.3	Construction of the objective function	13
3.1.4	Dynamic optimization under uncertainty	13
3.1.5	Model evaluation	13
3.2	Data Collection	15
3.3	Ethics	16
3.4	Limitations and Delimitations	17

4	Model Construction	18
4.1	Working Definition of Financial Well-Being	18
4.2	Dynamic Optimization Model	19
4.2.1	Model setup and overview	19
4.2.2	Life events	20
4.2.3	Income and expenses	22
4.2.4	Net cash flow	24
4.2.5	State variables	24
4.2.6	Decision variables and budget constraint	25
4.2.7	Buffer requirement	27
4.2.8	Down payment goal	27
4.2.9	Short-term savings	28
4.2.10	State transitions	29
4.2.11	Utility functions	30
4.2.12	Bellman equation	32
4.2.13	Model parameters	33
4.2.14	Python implementation	33
5	Results and Analysis	35
5.1	Overview	35
5.2	Comparison of the Utility Functions	36
5.3	Comparison of Beta Specifications	39
5.4	Evaluating the House Down Payment Bonus	40
5.5	Assessing Different Scenarios	41
5.5.1	Performance when the housing target becomes more demanding	41
5.5.2	Higher income scenario	41
5.5.3	Constrained income scenario	42
5.5.4	Constrained income combined with a demanding housing target	43
5.6	Sensitivity Analysis	43
5.7	Comparison of the Strongest Candidates	44
6	Discussion	45
6.1	Interpretation of the Main Results	45
6.2	Implications for Financial Advisory	45
6.3	Implementation in Financial Advisory	46
6.4	Validity	47
6.5	Reliability	48

7	Conclusions and Future Work	50
7.1	Conclusions	50
7.2	Future Work	51
	References	53
A	Simulation Results	57
A.1	Baseline scenario without housing bonus	57
A.2	Baseline scenario with housing bonus	58
A.3	Higher income and demanding housing target without housing bonus	58
A.4	Higher income and demanding housing target with housing bonus .	59
A.5	Higher income and highly demanding housing target without housing bonus	59
A.6	Constrained income scenario without housing bonus	60
A.7	Constrained income scenario with housing bonus	60
A.8	Constrained income and demanding housing target with housing bonus	61
A.9	Demanding housing target with reduced housing bonus	61
A.10	Demanding housing target with low housing bonus	61
A.11	Demanding housing target with increased housing bonus	62
A.12	Demanding housing target with high housing bonus	62
A.13	Demanding housing target with very high housing bonus	62
A.14	Demanding housing target without housing bonus	62
A.15	Constrained income scenario with reduced housing bonus	63
A.16	Constrained income scenario with increased housing bonus	63
A.17	Higher income and demanding housing target with increased housing bonus	63
A.18	Higher income and demanding housing target with reduced housing bonus	64

Chapter 1

Introduction

1.1 Background

Most people do not approach financial decisions in a structured way, but the consequences quickly become clear when planning falls short. Insufficient buffers, delayed housing entry, and inadequate long-term savings can quickly translate into financial stress and reduced resilience to economic shocks [1]. In practice, individuals must continuously balance competing goals such as maintaining liquidity, saving for major purchases, and building long-term wealth. These trade-offs are further complicated by uncertainty in income, expenses, and life events, making static financial advice insufficient for capturing the dynamic nature of household decision-making.

In recent years, there has been a growing focus among financial institutions and policymakers on financial well-being as a way to capture overall economic stability [1]. Rather than focusing solely on risk and return, financial well-being reflects an individual's ability to meet short-term obligations while maintaining resilience and progressing toward long-term goals [2]. At the same time, households differ depending on individual and household needs, priorities, and circumstances, suggesting that standardized financial advice may not be effective for all individuals [3]. This creates a need for structured and scalable frameworks that can support more personalized and transparent financial decision-making.

This thesis is conducted in collaboration with Handelsbanken and focuses on individuals aged 18–30. For this group, financial planning often involves balancing several important trade-offs, including housing entry, liquidity, and long-term saving. In Sweden, entering the housing market typically requires a substantial down payment, which constitutes a major financial barrier [4]. Alongside this, individuals need to maintain a sufficient level of liquid assets in order to buffer against income uncertainty,

while also saving for future consumption [5]. Despite the importance of these decisions, financial advisory services face constraints in terms of scalability and the ability to tailor recommendations to individual circumstances [6].

From a research perspective, these challenges can be viewed as a dynamic optimization problem under uncertainty [7]. This motivates the use of dynamic programming, since the individual's financial decisions are sequential and current allocations affect future financial outcomes.

Building on this, the thesis connects dynamic optimization with previous research on financial well-being, dynamic financial decision-making and goal-based wealth management [8, 2], but applies it in a banking advisory context. Instead of focusing only on investment portfolios or long-term wealth accumulation, this study considers several measurable components of financial well-being, including buffer savings, housing savings, short-term savings, and long-term savings.

1.2 Problem

Individuals often face several competing financial objectives at the same time [8]. For example, they need to maintain sufficient liquidity for unexpected expenses, save toward a housing down payment, finance short- and medium-term consumption goals, and build long-term financial stability. These objectives are interdependent, since allocating more resources to one goal reduces the resources available for others.

For the target group considered in this thesis, these trade-offs are especially important. Salary statistics by age indicate that income levels vary over the life cycle [9]. Younger adults are often at an early stage of their financial lives, where income, savings, and financial stability are still developing, while also facing major financial decisions such as saving for housing. This can result in limited financial margins and a greater need to balance short-term liquidity with long-term goals.

At the same time, research indicates that financial literacy and confidence are often lower among younger populations, which can make financial decisions more difficult to evaluate and act upon [10]. In a banking advisory context, this increases the need for guidance that is clear, individualized, and easy to communicate. However, such advice must also be structured and scalable enough to be used in practice. Existing advisory approaches often treat goals separately or rely on static rules of thumb, which is valuable because it is simple, transparent, and practically available. Simultaneously, they are limited in their ability to analyze how financial resources should be allocated dynamically across several competing goals under uncertainty.

The scientific problem is therefore how financial well-being can be represented in a quantitative model that captures several competing goals simultaneously. More specifically, the challenge is to construct a measurable index of financial well-being

and integrate it into a dynamic optimization framework that accounts for uncertainty in income, expenses, unemployment, and financial shocks.

1.3 Purpose

The purpose of this study is to contribute to more individualized and transparent financial advisory by developing a structured optimization framework for financial well-being in a Swedish banking context. From an academic perspective, the thesis aims to examine how financial well-being can be operationalized as a composite index and incorporated into a dynamic programming model. From an industrial perspective, the thesis explores how such a model could support banks in analyzing trade-offs between competing customer goals in a scalable way.

The model should also be understood as normative rather than descriptive. Its purpose is not to reproduce how individuals actually behave in all situations, but rather as a guide to how financial resources could be allocated over time in order to support financial well-being.

1.4 Research Question

The main research question is:

How can a composite index of financial well-being be constructed and integrated into a dynamic optimization framework that produces reasonable allocation behavior under exogenous risks?

To answer this question, the thesis addresses the following sub-questions:

1. How should financial well-being be defined and delimited in the context of individual banking advice?
2. Which measurable financial dimensions should be included in a composite index of financial well-being?
3. How can these dimensions be aggregated into an objective function suitable for dynamic optimization?
4. How do exogenous risks affect the optimal allocation between competing financial goals over time?
5. How sensitive is the model's allocation behavior to alternative parameter values and modeling assumptions?

1.5 Research Methodology

This thesis follows a quantitative modeling approach. First, financial well-being is operationalized into measurable components based on literature, banking advisory practice, and discussions with supervisors at Handelsbanken. Second, these components are incorporated into a utility function and embedded in a dynamic programming model. Third, the model is solved numerically in Python and evaluated using Monte Carlo simulation and sensitivity analysis under different scenarios.

The detailed model formulation, assumptions, and evaluation procedure are presented in Chapter 3.

1.6 Disposition

Chapter 2 presents the theoretical background of the thesis, including dynamic programming, Monte Carlo simulation, and the literature review. Chapter 3 describes the research methodology, covering the research process, data collection, ethics, limitations, and delimitations. Chapter 4 presents the model construction, including the working definition of financial well-being and the dynamic optimization model. Chapter 5 presents the results and analysis, focusing on the comparison of utility specifications, beta specifications, the housing bonus, scenario outcomes, sensitivity analysis, and the strongest model candidates. Chapter 6 discusses the main findings, their implications for financial advisory, practical implementation, validity, and reliability. Finally, Chapter 7 summarizes the thesis, and presents conclusions and suggestions for future work.

Chapter 2

Background

This chapter presents the theoretical background of the thesis, including dynamic programming, Monte Carlo simulation, and the literature review.

2.1 Dynamic Programming

Dynamic programming is a method for solving multi-stage decision problems in which current decisions affect both present outcomes and future opportunities. The central idea is to decompose a complex decision problem into a sequence of smaller subproblems, each associated with a particular stage and state. Richard E. Bellman formalized this approach by showing how such problems can be solved recursively, with the solution to each subproblem contributing to the overall optimum [7, p. 83].

A key concept in dynamic programming is the principle of optimality, which states that an optimal policy must have the property that, regardless of the initial state and decision, the remaining decisions must themselves constitute an optimal policy for the state resulting from the first decision [7, p. 83]. This principle makes it possible to formulate the problem recursively through a value function, which represents the maximum attainable value from any given state onward. The value function is characterized by the Bellman equation, which combines the current reward or cost with the continuation value of future states. The Bellman equation can be written as

$$V(s) = \max_{a \in A(s)} \left\{ R(s, a) + \gamma \sum_{s'} P(s' | s, a) V(s') \right\}, \quad (1)$$

where s denotes the current state of the system, and a denotes a decision or action chosen from the feasible action set $A(s)$. The term $R(s, a)$ represents the immediate reward obtained from choosing action a in state s . The transition probability $P(s' |$

s, a) describes the probability of moving to a future state s' given the current state and action. The function $V(s')$ represents the value of the future state, and γ is a discount factor that determines how future rewards are weighted relative to current rewards. In this thesis, the same recursive logic is used to evaluate financial allocation decisions over time, where the state represents the individual's financial position and the actions represent allocation choices between competing financial goals. Dynamic programming therefore not only identifies an optimal path, but also identifies an optimal decision policy for every possible state that may arise [11, pp. 20–27].

The computational complexity of dynamic programming is closely related to the size of the state space. As more state variables are introduced, the number of possible states increases rapidly, which can make the problem difficult to solve in practice [11]. The issue is particularly relevant in life-cycle and household finance models, where several dimensions such as household composition and living situation must be considered simultaneously. For instance, including factors like marital status and the presence of children requires a higher-dimensional state space, which increases computational demands and may limit the feasibility of exact solutions. Hence, a clear delimitation of the state variables included in the model is necessary to ensure that it remains computationally tractable in terms of running time.

2.2 Monte Carlo Simulation

Monte Carlo simulation is a method for analyzing systems affected by uncertainty through repeated random sampling. Instead of producing a single deterministic outcome, the method generates a distribution of possible outcomes, which makes it useful for evaluating risk, variability, and robustness [12].

In this thesis, Monte Carlo simulation is used to evaluate the policy generated by the dynamic programming model. After the optimal policy has been computed, the system is simulated forward under repeated realizations of uncertainty, including income variation, life events, and short-term spending. This approach follows standard practice in stochastic dynamic programming, where simulation is used to assess the performance of a policy under uncertainty [11].

The simulation results allow for the analysis of outcome distributions, such as goal attainment, variability, and robustness across different scenarios. However, Monte Carlo simulation does not determine whether the underlying model assumptions are realistic. Rather, it evaluates the implications of those assumptions, and the results should therefore be interpreted together with the model's assumptions, parameter choices, and sensitivity analysis.

2.3 Literature Review

This section reviews the research areas that motivate the modeling choices in this thesis. The literature is divided into five areas: financial advice for young adults, financial well-being measurement, goal-based wealth management, financial crisis as an exogenous risk, and precautionary saving. Together, these streams motivate the need for a dynamic optimization framework that can represent several financial goals, incorporate uncertainty, and translate financial well-being into measurable model components.

2.3.1 Financial advice for young adults

Sundarasan et al. highlight younger adults as a relevant group in financial behavior research, since this life stage often involves a transition from financial dependence to independent financial decision-making [13]. During this period, individuals begin to take greater responsibility for decisions related to debt, savings, investments, and long-term financial planning. This makes the ability to understand and evaluate financial choices particularly important for the target group.

Empirical evidence also suggests that younger individuals often have lower financial literacy and lower confidence in managing their personal finances. A Lund University study finds that many young adults struggle with basic financial concepts such as interest rates and inflation, and that confidence in financial decision-making is limited among parts of this group [10]. This indicates that younger adults may face difficulties when evaluating financial trade-offs on their own.

For the present thesis, this literature motivates the focus on individuals aged 18–30 and the need for advisory tools that are clear, structured, and easy to communicate. Since the target group may face several simultaneous financial goals while also having limited financial confidence, a model-based framework can help make allocation trade-offs more transparent.

In addition to the behavioral and financial literacy motivations, the choice to focus on younger individuals is also methodologically motivated. Compared to older households, younger adults typically face fewer complex life events, e.g., marriage, divorce, or childbearing, that significantly affect financial constraints and preferences over time.

This is relevant from a modeling perspective. In the life-cycle household finance literature, such as Bajari et al., individuals are modeled in environments where household composition evolves over time [14]. This requires accounting for transitions in marital status, fertility, and joint decision-making, which introduce additional state variables and increase model complexity.

By contrast, younger individuals are more likely to be single, without children, and solely responsible for their own financial decisions. Their decision problems can therefore be described using fewer state variables, which makes the model more tractable. This simplification is particularly useful when developing advisory tools, as it allows for an efficient and more easily interpretable baseline model.

2.3.2 Financial well-being measurement

A further relevant contribution is the work by the Consumer Financial Protection Bureau (CFPB) on measuring financial well-being [2]. The CFPB develops a financial well-being scale based on consumer responses, with the purpose of measuring how individuals experience their financial situation [2]. The scale captures several dimensions of financial well-being, including control over day-to-day finances, the ability to absorb financial shocks, progress toward financial goals, and financial freedom to make choices.

This work is relevant for the present thesis because it shows that financial well-being is a multidimensional concept rather than a single measure of wealth or income. The CFPB approach is primarily survey-based and captures subjective perceptions of financial well-being. Even though this is an important aspect of financial well-being, due to project limitations such as time and data collection, our study focuses on a quantifiable aspect. Therefore, we use quantitative data that is more easily translated into observable financial components of an optimization model.

2.3.3 Goal-based wealth management

Goal-based wealth management provides a useful reference point for this thesis because it studies financial decision-making as a problem of allocating limited resources across several goals over time. Das et al. develop a dynamic programming framework for multi-goal wealth management, where an investor allocates resources across goals such as housing, education, and retirement [8]. Their work illustrates that dynamic programming can be used to model sequential financial decisions when several objectives compete for the same resources.

However, the setting in this thesis differs from traditional goal-based wealth management. Das et al. primarily focus on investment allocation for predefined wealth goals, whereas this thesis applies a similar optimization logic to individual financial well-being in a banking advisory context. This means that the model not only considers long-term wealth accumulation, but also short-term liquidity, emergency buffer savings, housing down payment savings, and short-term savings.

For the present thesis, this literature motivates the use of dynamic programming

as the main modeling framework. Since financial decisions are sequential and current allocations affect future financial states, dynamic programming provides a suitable method for deriving allocation rules across competing goals over time.

2.3.4 Financial crisis as an exogenous risk

Exogenous risks are central in household decision-making, since individuals make saving and allocation decisions without knowing future income, employment status, or macroeconomic conditions with certainty. In dynamic financial models, such risks are commonly represented as stochastic processes or discrete state transitions that affect future resources and constraints.

A natural starting point is to consider how such risks are incorporated into household decision-making models. Bajari et al. develop a dynamic model of housing demand in which households face uncertainty in both future income and housing prices [14]. These variables are modeled as stochastic processes, reflecting the inability of individuals to predict future economic conditions. Their framework is partly motivated by events such as the global financial crisis of 2007–2008, during which households experienced large declines in housing prices, negative income shocks, and tighter credit conditions. This highlights how macroeconomic shocks can significantly influence both saving behavior and the timing of major financial decisions.

While Bajari et al. demonstrate the importance of incorporating such shocks, an additional question is how frequently they occur. Greenwood et al. address this by modeling financial crises as stochastic events and estimate that, at any point in time, the probability of a financial crisis occurring within the next three years is approximately 7% in normal times [15]. Additionally, empirical evidence further supports the magnitude of their impact at the household level. Jensen and Johannesen analyze the 2007–2008 financial crisis in a Danish context and show that macroeconomic shocks can have substantial effects on household consumption and financial positions [16]. Their results indicate that net worth declines significantly in the event of a financial crisis, demonstrating that such risks, while rare, have tangible and economically meaningful consequences for individuals. Altogether, this motivates modeling exogenous risks since they have real consequences at the household level.

Taken together, this literature provides a motivation for modeling financial crises as exogenous stochastic events. Bajari et al. motivate their inclusion in dynamic decision-making frameworks, Greenwood et al. provide evidence on their likelihood, and Jensen and Johannesen highlight their impact on household outcomes. Incorporating such risks is therefore essential for capturing realistic financial behavior,

particularly when individuals must balance short-term resilience against long-term financial goals.

2.3.5 Precautionary saving and buffer-stock models

Precautionary saving and buffer-stock models are also relevant to this thesis. Carroll argues that household saving behavior can be better described by a buffer-stock model than by the traditional life-cycle/permanent-income framework [5]. In this setting, individuals facing income uncertainty hold liquid wealth as a buffer against adverse income shocks. When wealth is below a desired target level, households have stronger incentives to save, while wealth above the target may allow for higher consumption.

This literature motivates treating buffer savings differently from other savings categories. In buffer-stock models, liquid wealth primarily serves to protect against income uncertainty rather than to maximize long-term returns. Accordingly, this thesis models the buffer as a target-based component, where the individual is encouraged to maintain sufficient liquidity rather than accumulate it indefinitely.

Furthermore, traditional buffer-stock models mainly focus on income uncertainty, consumption, and liquid wealth. This thesis extends the same precautionary saving logic to a multi-goal setting, where the buffer must be balanced against housing savings, short-term savings, and long-term savings within the same optimization framework.

2.4 Summary

Table 2.1 provides an overview of the research areas discussed in this chapter and their relevance to the thesis. It summarizes the focus, strengths, and limitations of each area, and shows how the present study combines these perspectives into a dynamic optimization framework for financial well-being under exogenous risk.

Research area	Focus	Strength	Limitation
Dynamic programming	Sequential decision-making where current decisions affect future outcomes	Provides a structured framework for deriving an optimal allocation policy	Becomes computationally demanding as number of dimensions increases
Monte Carlo simulation	Evaluation of outcomes under exogenous risk	Enables analysis of variability and robustness of the generated policy	Does not validate whether the underlying assumptions are realistic
Financial advice for young adults	Financial decision-making under low financial experience and early independence	Motivates the need for clear, structured, and interpretable advisory tools for the target group	Simplification for usability comes at the cost of capturing more complex life-cycle dynamics
Financial well-being measurement	Defining and measuring financial well-being as a multidimensional concept	Shows that financial well-being includes liquidity, resilience, and goal progress rather than only wealth or income	Subjective measures from survey-based studies are often included
Goal-based wealth management	Allocation of limited resources across multiple financial goals over time	Demonstrates how dynamic programming can be used to handle competing financial objectives	Primarily focuses on resource allocation and predefined goals rather than broader financial well-being
Financial crisis as an exogenous risk	Macroeconomic shocks affecting income and net worth	Motivates the inclusion of rare but economically relevant external risks in the model	Crisis probability and effects are difficult to estimate and may vary across contexts
Precautionary saving and buffer-stock models	Precautionary saving under income uncertainty	Captures liquidity and resilience	Limited to consumption and liquid wealth
This thesis	Multi-goal optimization of financial well-being under exogenous risks	Combines dynamic optimization, goal-based saving, liquidity, and uncertainty in a unified framework	Relies on simplifying assumptions and a limited state space to remain computationally tractable

Table 2.1: Summary of related research areas and their relation to this thesis

Chapter 3

Methodology

This chapter describes the research methodology, covering the research process, data collection, ethics, limitations, and delimitations.

3.1 Research Process

The methodology follows a quantitative modeling approach and the research process is organized around the research questions stated in Chapter 1. First, financial well-being is defined and delimited for the banking advisory context. Second, the financial dimensions included in the composite index are selected. Third, these dimensions are translated into an objective function that can be used in the optimization model. Finally, the resulting model is evaluated through simulation and sensitivity analysis in order to assess whether it produces reasonable allocation behavior under exogenous risks.

3.1.1 Definition of financial well-being

The first part of the research process addresses RQ1. Financial well-being is defined and delimited through a pre-study combining relevant literature, banking advisory practice, and discussions with supervisors at Handelsbanken. The purpose of this step is to translate a broad concept into an operational definition that can be used in a mathematical model. The outcome is a definition of financial well-being based on measurable financial goals relevant to the target group and advisory context.

3.1.2 Selection of financial dimensions

The second part addresses RQ2. Based on the operational definition, the financial dimensions included in the composite index are selected. These dimensions are chosen to reflect the main financial trade-offs faced by the target group, which are maintaining an emergency buffer, saving for a housing down payment, building short-term savings, and accumulating long-term savings. These dimensions later form the basis for the state variables in the dynamic optimization model.

3.1.3 Construction of the objective function

The third part addresses RQ3. The selected financial dimensions are aggregated into an objective function through a period utility function. Each component is assigned a functional form that reflects its economic role in the model. For example, buffer savings are treated as a target-based component, while other savings categories are represented through utility terms that capture progress over time.

Since the formulation of the utility function affects how the model prioritizes between financial goals, two alternative utility specifications are developed and compared. Comparing these alternatives makes it possible to assess which formulation produces more reasonable allocation behavior in the model. The detailed mathematical formulation of the utility functions, state variables, constraints, and transitions is presented in Chapter 4.

3.1.4 Dynamic optimization under uncertainty

The fourth part addresses RQ4. The chosen objective function is embedded into a dynamic programming model where the individual's financial situation is represented as a state and decisions are made sequentially over time. Uncertainty is introduced through stochastic income, expenses, life events, and short-term spending. The model is solved numerically in Python for a given set of parameter values, resulting in an optimal policy function. This policy function specifies how resources should be allocated across the financial components for each possible state in the discretized state space.

3.1.5 Model evaluation

The evaluation connects the model construction to RQ4 and RQ5. RQ4 is evaluated by studying how the optimal policy allocates resources under exogenous risks, while RQ5 is evaluated by comparing outcomes across alternative parameter values and model

assumptions. The evaluation is based on Monte Carlo simulation and sensitivity analysis.

Monte Carlo simulation

Monte Carlo simulation is used to evaluate the optimal policy under repeated realizations of uncertainty. For each specification, the model generates stochastic trajectories of income, expenses, life events, and short-term spending. The resulting outcomes are used to assess how the policy performs across possible future scenarios.

The main evaluation metrics are the success rate of reaching the down payment goal, the average first hitting time for the down payment goal, the average buffer level, the time spent below the critical buffer threshold, and the distribution of savings across the financial components. Each specification is evaluated using 5,000 Monte Carlo simulations, which allows the analysis to be based on distributions of outcomes rather than single simulated paths.

Sensitivity analysis

Sensitivity analysis is used to address RQ5. Since the composite index depends on parameter choices, the analysis focuses on how the results change when key parameters are varied. Particular attention is given to the utility weights β_i and the housing-goal bonus b , since these parameters determine how strongly the model prioritizes different financial goals. The sensitivity analysis is conducted as follows:

1. Set initial values for the parameter vector $\bar{\beta}$ and the bonus parameter b .
2. Solve the dynamic programming model and obtain the optimal policy for each state.
3. Run a Monte Carlo simulation based on this policy to generate stochastic outcomes.
4. Repeat the procedure for alternative parameter values and scenarios.
5. Compare the results across specifications to evaluate how parameter choices affect model outcomes.

This procedure makes it possible to analyze whether the model's conclusions are stable across scenarios and alternative assumptions or whether they depend strongly on specific parameter choices.

Scenarios used for evaluating robustness

The scenarios include a more demanding housing-goal scenario, a higher-income scenario, and a constrained-income scenario. The more demanding housing-goal scenario is represented by increasing the housing price from the baseline level of 1.5 MSEK to 2.5 MSEK. More generally, the model is tested on these housing price levels since they are plausible benchmark values for a first home, depending on the Swedish region considered [17]. The higher-income scenario is set to 35 KSEK per month in net income, as this represents a financially stronger but still realistic income level for the target group. It is therefore used to examine how the model behaves when the individual has greater saving capacity. To model a constrained-income scenario, we assume a conservative monthly income of 17 KSEK, based on wage levels observed in low-paid occupations such as cleaning [18]. Since Sweden does not have a statutory minimum wage, this value should be interpreted as a scenario assumption rather than an official minimum income. Nevertheless, the scenario is relevant for evaluating how the model prioritizes between competing financial goals when financial margins are limited.

Evaluation criteria

The model specifications are evaluated on whether they produce outcomes that are both financially sound and consistent with the definition of financial well-being used in the thesis. A specification is considered robust if it produces intuitive resource allocation behavior. In this context, intuitive means prioritizing buffer savings and progress toward the housing down payment, while still maintaining some allocation to short- and long-term savings. This reflects a reasonable balance between goal attainment and financial stability.

More specifically, the evaluation considers whether the model reaches the down payment goal within a reasonable time, maintains an acceptable buffer level, avoids excessive time below the critical buffer threshold, and produces policy shares that are financially interpretable.

3.2 Data Collection

The data used in this thesis were collected throughout the model construction process and consists of a combination of publicly available statistics, market assumptions, internal banking information, and academic literature. Rather than estimating the model empirically, the data is mainly used to calibrate parameters and create realistic scenario assumptions.

Publicly available data were used for several key model inputs. Income-related parameters, including historical income growth and unemployment statistics, were collected from Statistics Sweden (SCB). Expenditure assumptions were based on The Swedish Consumer Agency's estimated household costs for the relevant age group. Additional macroeconomic and institutional assumptions, such as inflation, unemployment benefits, and housing-market conditions, were collected from publicly available sources and official institutions.

In addition to public statistical data, Handelsbanken was used as an important source of banking-specific assumptions and contextual validation. Public information from Handelsbanken, such as the down payment requirement and savings-related return assumptions, was used to support the model calibration. Handelsbanken also provided internal contextual insights, including income-related spending patterns and discussions with supervisors regarding reasonable parameter choices and the practical interpretation of savings goals in an advisory setting. It should be noted that all internal information was used only at an aggregated level, and no individual customer data was included in the model.

Overall, the data collection should therefore be understood as a calibration process. The collected data and assumptions were used to construct a stylized but realistic model environment in which the dynamic programming framework could be tested. The specific data sources and parameter choices are described in more detail in Chapter 4, where they are introduced in connection with the corresponding model components.

3.3 Ethics

To gain access to Handelsbanken's customer transaction data, and to comply with regulations such as the General Data Protection Regulation (GDPR) and the bank's internal confidentiality agreements (NDAs), we were required to sign a formal data access and confidentiality contract. Only aggregated and fully anonymized results may be exported for academic purposes, and these exports are subject to internal approval procedures by our supervisors at the bank. Hence, we comply with GDPR regulations and ensure that only data strictly necessary for the research objective is accessed. Furthermore, it is standard that all students who conduct thesis projects within the bank receive a small compensatory payment.

Generative AI tools, such as ChatGPT, have been used as support during the project, primarily for code generation in Python and for improving the language and structure of the written report.

3.4 Limitations and Delimitations

A key limitation of the present study is that financial well-being is inherently a broad concept that cannot be fully captured within a mathematical optimization model. The chosen formulation reflects a deliberate simplification where financial well-being is restricted to a set of measurable financial goals that are relevant for the target group and the advisory context. As a result, subjective aspects such as perceived financial stress, individual preferences, and behavioral biases are not explicitly modeled. Similarly, only a limited number of life events are included, since incorporating every event that affects individual finances during a lifetime is not feasible.

Additionally, the study is subject to several delimitations. First, the analysis is restricted to a younger target group (ages 18 to 30), which means that the model is not intended to represent the financial circumstances of the population as a whole. Second, our chosen exogenous risks are unemployment and financial crisis, which are relevant for our chosen target group. Hence, our approach builds a baseline model that illustrates how exogenous risk factors can be incorporated into the modeling framework, while leaving room for future extensions. Further delimitations such as assumptions, simplifications, and modeling choices are introduced in the detailed description of the optimization model and its construction, see Chapter 4.

Chapter 4

Model Construction

This chapter presents the model construction, including the working definition of financial well-being and the dynamic optimization model.

4.1 Working Definition of Financial Well-Being

Financial well-being is a broad concept that has many definitions and can include several different dimensions. In short, it can be understood as the state in which individuals are able to meet current financial obligations and absorb financial shocks without compromising future opportunities [1]. However, such a broad definition is difficult to use directly in a mathematical optimization model. For this reason, a pre-study was conducted in order to operationalize the concept into measurable components. In particular, financial well-being is interpreted here in the context of individual financial planning and banking advisory services. Rather than attempting to capture psychological and subjective aspects, the study constructs a practical representation that reflects selected financially relevant goals. This approach is partly based on existing literature, but also on discussions with supervisors at Handelsbanken [1, 6, 19, 20]. In this thesis, the composite index of financial well-being is defined as follows:

1. **Buffer:** Build and maintain an emergency buffer corresponding to approximately two months of net salary [21].
2. **Down payment:** Save for a housing down payment for a first home. Although the current minimum down payment requirement is 10%, the model uses 15% as a conservative baseline assumption [4, 22].

3. **Short-term savings goal:** Save toward a short-term goal within 1–5 years (e.g. travel, a car, or home-related purchases). This goal aims to capture consumption above the defined minimum spending level [6].
4. **Long-term savings goal:** Save toward long-term objectives beyond five years, such as pension or wealth accumulation [6].

The idea is that financial well-being depends on how well an individual balances these components over time, given limited resources and exposure to exogenous risks. A higher level of financial well-being is associated with maintaining a sufficient buffer, making progress toward the housing goal, and accumulating long-term savings, while still allowing for consumption above a minimum level.

4.2 Dynamic Optimization Model

4.2.1 Model setup and overview

The model is formulated for individuals aged 18–30 who do not currently own a home and aim to purchase their first home within a finite planning horizon. Each individual is characterized by observable background characteristics, denoted by Z . These characteristics are treated as fixed model inputs and affect income, expenses, and housing-price assumptions.

Time is modeled in annual periods in order to reduce the number of decision stages and keep the state space computationally tractable. This is also consistent with the financial goals considered in the model, since goals such as housing savings and long-term savings are typically achieved over several years rather than within a few months. Similarly, banking advice is generally oriented toward medium- and long-term financial planning, rather than detailed recommendations for exact allocations over short monthly intervals. Within each annual period t , events occur in the following order:

1. The current state is observed, i.e., whether the individual is in a normal state, unemployment state, or financial crisis state.
2. Income and expenses are realized.
3. Net cash flow is determined.
4. Saving decisions are made based on the available net cash flow.
5. Stochastic short-term spending is realized.

6. The financial state transitions to the next period according to the transition equations.

4.2.2 Life events

The individual is assumed to be in one of three possible life-event states at the beginning of each period, t :

$$\chi_t \in \{N, FC, U\}.$$

Here, N denotes a normal period without financial disruption, FC denotes a financial crisis, and U denotes unemployment. In this context, a financial crisis mainly decreases an individual's assets, while unemployment affects the income.

As a simplifying assumption, only one life event can occur each year. Moreover, a normal state is always assumed to follow a financial crisis before another shock state can occur. Similarly, following an unemployment state, the person is either still unemployed the next year or in a normal state. Hence, the model does not allow direct transitions from financial crisis to unemployment or from unemployment to financial crisis. This restriction is introduced to keep the model tractable.

Financial crisis

Financial crises are modeled as an exogenous risk in the framework, following the literature review. As shown by Greenwood et al. [15], financial crises can be interpreted as stochastic events, with an estimated probability of approximately 7% of occurring within a three-year horizon. Since the model operates on an annual time step, this gives an annual probability, p , of approximately 2.4 percent, derived from:

$$1 - (1 - p)^3 = 0.07$$

To capture the economic consequences of such events at the household level, we draw on empirical evidence from Jensen and Johannesen [16] who analyzed the effects of the financial crisis in 2007–2008 on Danish households. The study found that the average individual's net worth declined by approximately 20,000 DKK, which we then calculated would correspond to roughly 38,900 SEK today [16, 23, 24].

Together, this allows the model to incorporate both the likelihood and the economic impact of financial crises, in line with the theoretical and empirical insights from the literature.

Unemployment

Another exogenous risk we chose to model is unemployment. It serves as an income risk that is particularly relevant for our target group, for which unemployment rates are substantially higher than the overall population. In 2025, the overall unemployment rate (ages 15-74) was 8.8 percent according to SCB's Labour Force Survey (AKU). For young individuals aged 15-24, SCB reports an unemployment rate of 24.3 percent in 2025 [25]. We represent unemployment risk using annual transition probabilities:

$$p_{NU} = \Pr(\chi_{t+1} = U \mid \chi_t = N), \quad p_{UN} = \Pr(\chi_{t+1} = N \mid \chi_t = U).$$

The probabilities are calibrated using an unemployment rate u and the average duration of unemployment d (months) from the SCB statistical database. Here, $d \approx 9.0$ months calculated from 2025 data based on age 20-65 [26]. Assuming a geometric exit process from unemployment, the monthly job-finding probability is approximated by $f \approx 1/d$, which implies an annual job-finding probability

$$p_{UN} = 1 - (1 - f)^{12}.$$

Thereafter, we approximate the dynamics as a two-state Markov chain between normal employment and unemployment. Note that financial crisis is an additional exogenous shock state with low probability. Hence, the calibration of unemployment risk should be interpreted as an approximation rather than as the exact stationary distribution of the full three-state process. This approach gives the steady-state relation:

$$u = \frac{p_{NU}}{p_{NU} + p_{UN}},$$

Here, $u = 0.088$ is the unemployment rate [25]. Following this, p_{NU} can be backed out given u and p_{UN} . Thus, the calibrated annual transition probabilities are approximately:

$$p_{UN} = 0.757, \quad p_{NU} = 0.073.$$

Note that the SCB statistics on the unemployment rate are not available for the exact target age group (18-30). We therefore use the overall unemployment rate of the population as a baseline. Furthermore, as the youth unemployment statistics also include many full-time students, the aggregated unemployment rate is better to use as a conservative baseline rather than a direct estimate for ages 18-30.

4.2.3 Income and expenses

The realized life event affects income and necessary expenses according to

$$I_t = I(Z, \chi_t), \quad E_t = E(Z).$$

Income growth rate

The nominal income growth rate, g_t , is defined as the relative change in income between two consecutive years

$$g_t = \frac{I_{t+1} - I_t}{I_t}.$$

Annual average income data for Swedish citizens between the years of 1992–2024 was obtained from SCB. The average annual income growth rate over this period is estimated to approximately $\bar{g} \approx 0.033$, corresponding to an average yearly increase of about 3.3%.

Income during unemployment and financial crisis

Unemployment is modeled as a temporary income reduction:

$$I_t = \begin{cases} I_t^N, & \chi_t \in \{N, FC\} \\ \rho \min(I_t^N, I^*), & \chi_t = U \end{cases}$$

where I_t is annual labor income. Income during unemployment is assumed to be partially replaced through the Swedish unemployment insurance system. In Sweden, unemployment benefits replace up to approximately 80 percent of previous earnings, subject to eligibility conditions and benefit ceilings [27]. We therefore set $\rho = 0.8$, where ρ denotes the income replacement rate. This is a simplification since effective replacement rates may be lower for some individuals. Following the Swedish Unemployment Insurance Inspectorate (IAF), the ceiling for the income base is 34 000 SEK per month, implying an annual ceiling of

$$I^* = 12 \times 34\,000.$$

Hence, if $I_t^N > I^*$, annual income is computed based on I^* rather than I_t^N .

Moreover, previous studies have shown that nominal incomes were not directly affected by a financial crisis but rather that the real income was affected [16]. To model this behavior, we assume that the income remains constant and equal to that during a normal year.

Expenses

Expenses are modeled as the sum of a minimum expenditure component and an income-linked expenditure component:

$$E_t = E_t^{\min}(Z) + E_t^{\text{income-linked}}.$$

Minimum expenses

The minimum expenditure component is based on The Swedish Consumer Agency's estimated household costs for the target group aged 18–30 reported in Table 4.1. The Swedish Consumer Agency presents these figures as estimates of how much a household may need to spend on essential items, rather than as observed expenditure data [28]. Following this, the yearly minimum expenses are 12 times the total in

Table 4.1: Baseline individual monthly expenses for the target group aged 18–30, based on The Swedish Consumer Agency [28]. The numbers are calculated as a weighted average of the age groups 18–24 and 25–50.

Category	SEK/month
Food	2,789
Clothing and shoes	847
Leisure	616
Mobile phone	130
Personal hygiene	579
Household	3,530
Total baseline individual expenses	8,491

the table, $E_t^{\min}(Z) = 101,892$. Note that household in the table refers to shared household costs for a one-person household, including consumables, household equipment, internet and mobile subscriptions, media services, electricity, water and sewage, and home insurance.

Income-linked spending

While minimum expenses provide a lower bound on necessary consumption, they do not capture how spending behavior changes as income increases. In practice, individuals do not keep consumption fixed at subsistence levels when their income rises. To capture this behavior, we introduce an income-linked expenditure component. This component reflects that part of spending that scales with income above the minimum level, such as improved housing conditions, higher-quality consumption, and increased discretionary spending.

An internal study from Handelsbanken provides empirical estimates of how additional expenditures vary with changes in income across income groups. The results show that fixed expenses increase by 450 SEK per month for each additional 1,000 SEK, for group 1 (net income 0–25 KSEK), and by 400 SEK for group 2 (net income of 25–50 KSEK). Following this, the income-linked expenditure component is defined as

$$E_t^{\text{income-linked}}(z) = \alpha_z \cdot \max \left\{ 0, \frac{I_t - E_t^{\min}(Z)}{1000} \right\}, \quad z \in \{1, 2\},$$

where

$$\alpha_1 = 450, \quad \alpha_2 = 400.$$

The minimum expenses are removed from the income to calculate the surplus left. Only income above this level gives rise to additional income-linked expenditures, which is captured by α_z that denotes the extra spending per 1,000 SEK excess.

Inflation assumption on expenses

We account for inflation using a constant annual rate of $\pi = 0.02$, which is the target annual rate for Sweden's central bank [29]. Consequently, nominal expenses are adjusted between periods by the inflation factor $(1 + \pi)$.

4.2.4 Net cash flow

After income and expenses are realized, the available net cash flow is defined as:

$$M_t = I_t - E_t.$$

All financial decisions in period t are made conditional on the realized net cash flow M_t . If $M_t > 0$, the surplus is allocated among the decision variables. If $M_t < 0$, there is a deficit, which will be financed by withdrawals from different savings.

4.2.5 State variables

The financial state at time t is defined by our working definition of financial well-being, that is:

$$S_t = (B_t, H_t, K_t, L_t),$$

where

- B_t denotes liquid buffer savings,

- H_t denotes savings for a housing down payment,
- K_t denotes short-term savings,
- L_t denotes long-term savings.

All state variables are restricted to be non-negative:

$$B_t, H_t, K_t, L_t \geq 0.$$

These state variables form the state space in the dynamic programming problem.

Initial Values

At the beginning of the planning horizon, the individual is assigned an initial financial state

$$S_0 = (B_0, H_0, K_0, L_0),$$

where S_0 denotes the initial levels of buffer savings, down payment savings, short-term savings, and long-term savings. These initial values are treated as fixed input parameters so that the same optimization framework can be applied to individuals with different financial starting points.

Discretization of the state space

In the theoretical model, the state variables B_t , H_t , K_t , and L_t are continuous. In the implemented dynamic programming model, however, the state space is approximated using a finite grid in order to make the problem computationally tractable. Since evaluating the value function at every possible continuous state would result in impractical run times, the value function is computed only at selected grid points.

The state grid is used when approximating the continuation value in the Bellman recursion. A coarse state grid is sufficient since the continuation value is evaluated over large financial state variables, while a finer grid would significantly increase computational cost.

4.2.6 Decision variables and budget constraint

Given the current financial state S_t and the realized net cash flow M_t , the individual chooses how to allocate resources across different financial components. These allocation choices are summarized by the decision vector x_t and determine how the current state evolves into the next period's state. It is defined as:

$$x_t = (s_t^B, s_t^H, s_t^K, s_t^L),$$

where

- s_t^B denotes the net allocation to buffer funds,
- s_t^H denotes the net allocation to the down payment,
- s_t^K denotes the net allocation to short-term funds,
- s_t^L denotes the net allocation to long-term funds.

The decisions must satisfy the budget constraint

$$s_t^B + s_t^H + s_t^K + s_t^L = M_t.$$

Furthermore, the sign of the decision variables must be consistent with the sign of net cash flow. When $M_t \geq 0$, all decision variables are weakly non-negative. When $M_t < 0$, all decision variables are weakly non-positive. Thus, positive net cash flow corresponds to new saving allocations, whereas negative net cash flow corresponds to withdrawals from previously accumulated funds. First, the deficit is covered by buffer savings, consistent with the purpose of the buffer as financial protection. Only if the buffer is insufficient is the remaining deficit taken from the other financial components.

Discretization of the decision space

To ensure computational tractability, the continuous decision space is approximated using a discrete action grid, where allocation shares are restricted to fixed increments of 25 percent of net cash flow. The decision space is discretized by the action grid, which is used to describe the possible allocation choices in each period. This grid is more detailed than the state grid, since the model needs to distinguish between different allocation shares across buffer savings, housing savings, short-term savings, and long-term savings. A finer action grid allows the policy to capture more nuanced saving decisions without expanding the full state space to the same extent.

As stated, allocations are restricted to increments of 25 percent of net cash flow. This discretization balances flexibility in allocation decisions with computational feasibility. Hence, in surplus periods, each decision share is chosen from

$$\{0, 0.25, 0.5, 0.75, 1.0\}.$$

This discretization defines the feasible set of allocation choices in each period, which forms the basis for evaluating how different decisions contribute to the individual's financial goals.

4.2.7 Buffer requirement

To ensure financial resilience for the consumer, an optimal buffer level, B^* , is set to two months' worth of income for the individual [21], that is:

$$B^*(Z) := \frac{1}{6}I_t.$$

4.2.8 Down payment goal

Let P_t denote the relevant housing price level, θ the required down payment ratio, and hmg the annual house-price market growth rate. In the optimization, $hmg = 0.03$, and the required down payment is defined as

$$H_t^{req} = \theta P_{\text{house}}(1 + hmg)^t.$$

The down payment goal is achieved when

$$H_t \geq H_t^{req}.$$

House prices are assumed to grow at a constant annual rate of 3 percent. This reflects a conservative long-run approximation, consistent with the historical upward trend in Swedish housing prices documented by Statistics Sweden (SCB) and international sources such as the Organisation for Economic Co-operation and Development [30, 31, 6]. The purpose of this component is not to model housing choice or characteristics, but rather to capture the financial threshold for entering the housing market. The housing price is therefore interpreted as an affordability benchmark.

As of 1 April 2026, the Swedish mortgage cap was raised from 85 to 90 percent of the market value of the home, which in practice lowers the minimum down payment requirement from 15 percent to 10 percent [22]. Nevertheless, the model retains $\theta = 0.15$ as a baseline assumption. This is a deliberate modeling choice, since first-time homebuyers typically face additional initial costs beyond the down payment itself, such as moving expenses, furnishings, and other housing-related purchases. Retaining the 15 percent threshold therefore gives the model some financial margin around the purchase decision.

The analysis is restricted to tenant-owned apartments. This delimitation is motivated by the chosen target group, namely individuals aged 18–30 who are assumed to be entering the housing market for the first time. SCB statistics on housing patterns by age indicate that apartments are a common housing form among younger adults, which makes them relevant to use for the present model [32]. At the same time, apartment prices differ substantially across regions, which motivates

allowing the housing target to vary across scenarios [17].

4.2.9 Short-term savings

Short-term savings represent funds allocated to expenditures and financial goals expected to arise within the next one to five years. Examples include travel, restaurant visits, education-related expenses, and other foreseeable purchases that require advance preparation but do not justify long-term investment.

This distinguishes short-term savings from long-term savings in an important way. While long-term savings are intended to remain invested throughout the planning horizon, short-term savings are instead built up in advance and intended to be used gradually for relevant expenditures. Therefore, short-term savings are not only modeled as an accumulating asset, but are also associated with a withdrawal variable. Let $w_t^K \geq 0$ denote the amount withdrawn from short-term savings in period t . This withdrawal represents the use of previously accumulated short-term funds for expenditures and captures the fact that funds are meant to finance needs that materialize and are realized within a limited period. The withdrawal is modeled as a proportion of the current stock of short-term savings:

$$w_t^K = \lambda_t K_t,$$

where K_t denotes the level of short-term savings in period t , and $\lambda_t \in [0, 1]$ is a stochastic withdrawal rate.

In the theoretical formulation, λ_t may be viewed as uniformly distributed on $[0, 1]$. However, in the dynamic programming implementation it is approximated using a discrete grid of possible spending levels with associated probabilities to help keep the Bellman equation computationally tractable. In contrast to this, in the Monte Carlo simulation, λ_t is drawn from a continuous uniform distribution on $[0, 1]$. Hence, the policy is optimized under a discrete approximation of spending, while its performance is evaluated under a continuous representation of spending.

Note that the spending is realized after the saving decision has been made in a given period. It therefore affects next period's asset levels through the state transition, rather than the policy choice within the same period.

4.2.10 State transitions

Given the real-world life event χ_t and the decisions $x_t = (s_t^B, s_t^H, s_t^K, s_t^L)$, the states of the competing goals evolve according to the following:

$$\begin{aligned} B_{t+1} &= B_t R^B + s_t^B, \\ H_{t+1} &= H_t R^H + s_t^H, \\ K_{t+1} &= K_t R^K + s_t^K - w_t^K, \\ L_{t+1} &= L_t R^L + s_t^L \end{aligned}$$

where each state variable evolves with a constant gross growth factor, denoted R^B, R^H, R^K, R^L . This defines the transition function used in the Bellman recursion.

Expected returns for each asset category

Expected returns for each financial component are implemented as constant annual gross growth factors. For each asset category i , returns are defined as $R^i = 1 + r^i$, where r^i denotes the expected annual return. The returns follow the return assumptions used in Handelsbanken's savings calculator, which links expected returns to investment horizon and risk level [33]. Funds allocated to long-term objectives do not require short-term liquidity and can therefore be invested in assets with higher risk and higher expected return. Accordingly, buffer savings are assigned $R^B = 1.01$ reflecting savings account returns, down payment savings $R^H = 1.02$ corresponding to low-risk fixed-income investments, short-term savings $R^K = 1.04$ representing balanced portfolios, and long-term savings $R^L = 1.07$ consistent with globally diversified equity portfolios.

State transition during financial crisis

In a financial crisis state, the additional crisis cost of 38 900 SEK is deducted after the ordinary state transition. This cost is first withdrawn from buffer savings. If buffer savings are not sufficient, the remaining amount is removed from the other savings categories through optimization. This representation should be interpreted as a reduced-form approximation of the household-level financial impact of a crisis, rather than a full macroeconomic crisis model.

4.2.11 Utility functions

To operationalize financial well-being in the dynamic programming model, the selected financial dimensions must be aggregated into a period utility function. Since the functional form determines how the model trades off liquidity, housing savings, short-term savings, and long-term savings, we compare two alternative utility specifications. The purpose of this comparison is not to estimate individual preferences, but to evaluate how different representations of financial well-being affect the resulting allocation behavior.

Both specifications include the same financial dimensions, but differ in how savings progress is measured. The first specification evaluates housing, short-term, and long-term savings in absolute monetary levels, while treating the buffer as a target-based component. The second specification evaluates savings relative to explicit target or reference levels, in order to test whether a target-relative formulation improves comparability across financial goals.

Alternative 1

Logarithmic utility is used to represent diminishing marginal utility of savings. This means that each additional unit of saving contributes less additional utility as the level of savings increases. The logarithmic form is therefore used as a modeling choice to capture concavity in the utility function. Financial well-being in period t is defined as

$$u_t(S_t) = -\beta_B \left(\frac{B_t - B_t^*}{B_t^* + \varepsilon} \right)^2 + \beta_H \ln(H_t + \varepsilon) + \mathbf{1}_{\{H_t \geq H_t^{\text{req}}\}} \cdot b + \beta_K \ln(K_t + \varepsilon) + \beta_L \ln(L_t + \varepsilon),$$

where $\varepsilon = 1$ is introduced to avoid $\ln(0)$, B_t^* denotes the target buffer level, b is a bonus term received when the down payment goal is reached, and the betas β_i are the utility weights. The utility weights β_i are treated as relative priority weights across the financial components. For comparability across specifications, the weights are normalized such that

$$\beta_B + \beta_H + \beta_K + \beta_L = 1.$$

The functional form also differs across the financial components. Logarithmic utility is used for housing savings, short-term savings, and long-term savings, whereas buffer savings are modeled as a target-based component. Rather than rewarding buffer accumulation indefinitely, the purpose of a buffer is to maintain sufficient liquidity. A quadratic penalty around the buffer target therefore reflects that both insufficient and excessive buffer amounts are less desirable than holding an amount close to the target level.

Furthermore, the housing-goal bonus b is treated as a separate parameter and is not included in the normalization. The bonus is added when the down payment goal is reached and is initially set to 0.5. The purpose of the bonus is to reflect that reaching the housing threshold has a value beyond the gradual increase captured by logarithmic utility on its own. Thus, the model recognizes that achieving the required down payment represents a discrete improvement in financial well-being, since it makes entry into the housing market feasible.

Alternative 2

The motivation for this specification is that absolute monetary levels may make the utility components difficult to compare directly. By expressing savings relative to target or reference levels, the formulation gives each component a clearer interpretation in terms of goal achievement and tests whether this improves the stability and interpretability of the resulting allocation behavior. Period utility is then defined as

$$u_t(S_t) = -\beta_B \left(\frac{B_t - B_t^*}{B_t^* + \varepsilon} \right)^2 + \beta_H \ln \left(\frac{H_t + \varepsilon}{H_t^* + \varepsilon} \right) + \beta_K \ln \left(\frac{K_t + \varepsilon}{K_t^* + \varepsilon} \right) + \beta_L \ln \left(\frac{L_t + \varepsilon}{L_t^* + \varepsilon} \right),$$

where H_t^* , K_t^* , and L_t^* denote the target levels for housing, short-term savings, and long-term savings, respectively. As in Alternative 1, buffer savings are modeled separately through a target-based penalty, since the purpose of the buffer is to maintain sufficient liquidity rather than to accumulate indefinitely. However, unlike Alternative 1, it does not include a separate housing bonus. Since housing savings are evaluated relative to the explicit target level H_t^* , progress toward the down payment goal is already incorporated in the utility function. The importance of this goal is therefore controlled through the weight β_H , rather than through an additional bonus term. This keeps the specification cleaner and allows us to test whether target-relative utility alone is sufficient to produce goal-oriented allocation behavior.

In contrast to the buffer and housing components, there is no equally well-defined external target for short-term and long-term savings. Rather, we introduce two reference levels in order to make these dimensions operational in the model. These levels should therefore be interpreted as modeling benchmarks rather than objectively optimal savings targets. The short-term reference level is defined as a rolling five-year benchmark,

$$K_t^* = \sum_{s=\max(0, t-4)}^t \kappa_K I_s^N (R^K)^{t-s},$$

where $\kappa_K = 0.05$ denotes that 5 percent of normal income is treated as the desired

annual contribution to short-term savings, I_s^N is normal income in period s , and R^K is the gross return on short-term savings. This formulation reflects that short-term savings are intended to finance expenditures within a limited future horizon, that is, one to five years. The long-term reference level is defined as

$$L_t^* = L_0^*(1 + \pi)^t,$$

where $L_0^* = 1,000,000$ SEK is the initial benchmark level for long-term savings and π is inflation. The values of κ_K and L_0^* are modelling assumptions used to construct reasonable reference levels for short-term and long-term savings. They were discussed with supervisors at Handelsbanken and considered plausible for the target group and advisory context [6].

4.2.12 Bellman equation

The components introduced above define the dynamic optimization problem. The state vector S_t describes the individual's financial position at the beginning of period t , the decision vector x_t determines how the realized net cash flow is allocated across financial goals, the transition equations describe how these decisions affect future financial states, and the utility function u_t evaluates the financial well-being associated with each state and decision.

Let $V_t(S_t)$ denote the value function. The value function represents the maximum expected financial well-being that can be achieved from period t onward, given the current financial state S_t . Using the Bellman equation (1), introduced in Section 2.1, the problem can be written recursively as

$$V_t(S_t) = \max_{x_t \in \mathcal{X}(S_t)} \{u_t(S_t, x_t) + \gamma \mathbb{E}[V_{t+1}(S_{t+1}) \mid S_t, x_t]\},$$

where $\mathcal{X}(S_t)$ denotes the feasible set of allocation decisions in state S_t , and $\gamma \in (0, 1)$ is the discount factor. The first term, $u_t(S_t, x_t)$, captures current-period financial well-being, while the second term captures the expected continuation value of future financial states. The expectation is taken over the stochastic elements in the model, including life events and short-term spending shocks.

In the numerical implementation, the discount factor is set to $\gamma = 0.95$, meaning that future financial well-being is valued slightly less than current financial well-being. Solving the Bellman recursion yields an optimal policy function that specifies how the individual should allocate available net cash flow across buffer savings, housing savings, short-term savings, and long-term savings for each possible financial state. This policy function forms the basis for the simulation and sensitivity analysis

presented in the following chapter.

4.2.13 Model parameters

The parameters presented in Table 4.2 summarize the key variables and parameter values used throughout the model. By this stage, all core components have been introduced, including the state space, decision variables, transition equations, and utility function. The table serves as a quick reference for the main parameter values and assumptions used in the dynamic programming model. These parameters determine the model environment, including transition probabilities, income and expense assumptions, return rates, savings targets, and utility-related inputs.

4.2.14 Python implementation

The model was implemented in Python using an object-oriented structure, where all model parameters are stored in a frozen dataclass and the dynamic programming logic is incorporated in a separate class. This class contains methods for computing income, expenses, net cash flow, feasible actions, state transitions, utility values, and life-event transition probabilities. The continuous state variables are approximated using predefined grids, and each state is mapped to the nearest grid point using a binary-search-based helper function. The action space is discretized by generating all feasible allocation-share vectors in 25 percent increments. In surplus periods, the code iterates over all possible allocations across the goals, while in deficit periods it applies a rule-based withdrawal order where the buffer is used first before withdrawals from the other components are considered.

The core optimization algorithm is implemented through a recursive value-function method, which evaluates all feasible actions for a given state and stores the action with the highest expected value. For each action, the code computes the current utility, simulates the resulting next financial state, discretizes it back onto the state grid, and then loops over all possible next life-event states and spending shocks to calculate the expected continuation value. To reduce repeated computations, the value function uses `lru_cache`, and the optimal actions are stored in separate policy dictionaries for later lookup. After solving the dynamic program, the resulting policy is evaluated using Monte Carlo simulation functions that repeatedly generate stochastic paths, apply the stored optimal policy, update the financial state over time, and summarize outcomes such as reaching the down-payment target, final savings levels, and average allocation shares.

Table 4.2: Model parameter values

Symbol	Description	Value
<i>Life events</i>		
p	Financial crisis probability	0.024
p_{NU}	Normal to unemployment	0.073
p_{UN}	Unemployment to normal	0.757
<i>Income and macro</i>		
I^*	Benefit income ceiling	408,000 SEK
ρ	Replacement rate	0.8
$\bar{g}$	Income growth	0.033
π	Inflation rate	0.02
<i>Expenses</i>		
$E_t^{\min}$	Minimum expenses	101,892 SEK
α_1	Spending parameter, group 1	450
α_2	Spending parameter, group 2	400
<i>Returns</i>		
R^B	Buffer return	1.01
R^H	Housing return	1.02
R^K	Short-term return	1.04
R^L	Long-term return	1.07
<i>Savings targets</i>		
λ_t	Withdrawal rate	$\lambda_t \sim U(0, 1)$
θ	Down payment ratio	0.15
hmg	Housing price growth	0.03
B_t^*	Buffer target	$\frac{1}{6}I_t$
κ_K	Short-term target share	0.05
L_0^*	Long-term reference level	1,000,000 SEK
<i>Utility</i>		
ϵ	Smoothing constant	1
b	Housing bonus	0.5
γ	Discount factor	Assumed
β_i	Utility weights	Varied

Chapter 5

Results and Analysis

This chapter presents the results and analysis, focusing on the comparison of utility specifications, beta specifications, the housing bonus, scenario outcomes, sensitivity analysis, and the strongest model candidates.

5.1 Overview

This chapter presents and analyzes the numerical results generated by the dynamic programming model. The analysis focuses on how different utility specifications, preference weights, housing targets, income levels, and housing-goal bonuses affect the optimal allocation of savings across buffer savings, housing savings, short-term savings, and long-term savings.

The evaluation is primarily based on whether the model reaches the down payment goal within the planning horizon, the average first hitting time of this goal, and the resulting balance between the remaining savings components. These metrics help to answer RQ5 by showing how the model responds to alternative parameter values and scenario assumptions. At the same time, they help answer RQ4 by illustrating whether the model can maintain financially interpretable allocation behavior across different economic scenarios under exogenous risk.

The chapter is structured as follows. First, the two utility formulations are compared in order to select the specification used in the remaining analysis. Second, different beta specifications are evaluated to identify allocation preferences that generate intuitive and robust saving behavior. Third, the effect of the housing-goal bonus is examined. Finally, the selected specifications are tested across different income and housing-price scenarios, followed by a sensitivity analysis of the bonus parameter.

5.2 Comparison of the Utility Functions

To evaluate the practical implications of the two formulations, we compare their simulated saving paths across selected runs. The purpose is not only to assess total wealth accumulation, but also to examine how each formulation prioritizes between down payment savings, short-term savings, and long-term savings.

The second formulation was introduced to test whether expressing savings relative to target or reference levels would make the utility components more comparable and easier to interpret in terms of goal achievement. As the results below indicate, the main difference between the two formulations is not a clear improvement in model performance, but rather a change in how resources are allocated across the financial goals. Note that for the figures below, blue represents buffer savings B_t , orange represents down payment savings H_t , green represents short-term savings K_t , red represents long-term savings L_t , and purple represents accumulated short-term consumption.

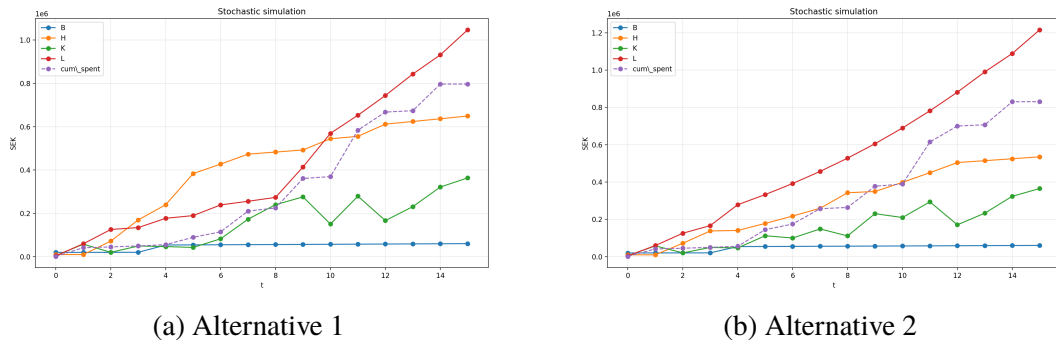
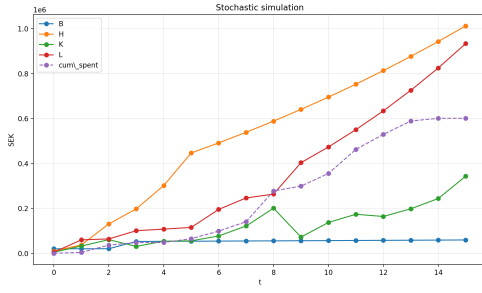
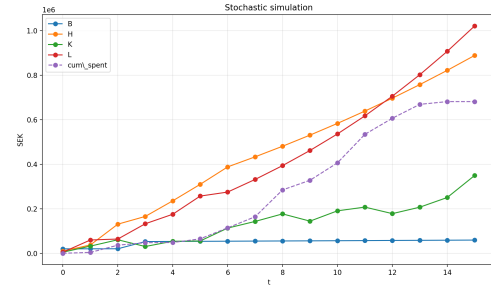


Figure 5.1: Comparison of utility functions when $\beta = (0.25, 0.25, 0.25, 0.25)$

Figure 5.1 is particularly informative because the beta weights are equal across all savings goals. Since no component is explicitly favored, the difference between the simulated paths can be attributed to the differing utility formulations. The figure suggests that Alternative 1 gives higher priority to H_t , whereas Alternative 2 allocates relatively more resources toward L_t . This difference indicates that the two formulations translate identical preference weights into different allocation policies. In this setting, the target-relative specification in Alternative 2 does not appear to create a sufficiently strong incentive to reach the housing target. By contrast, the housing bonus in Alternative 1 seems to reinforce goal-oriented behavior toward the down payment objective.



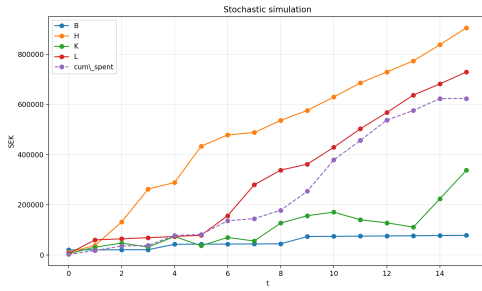
(a) Alternative 1



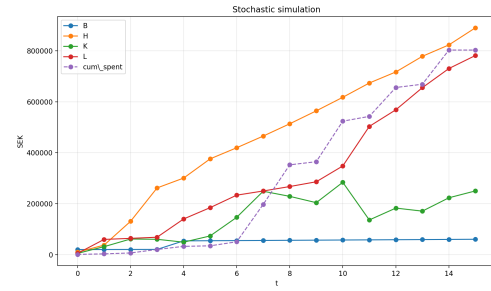
(b) Alternative 2

Figure 5.2: Comparison between utility functions when $\beta = (0.3, 0.4, 0.1, 0.2)$

Figure 5.2 reinforces this interpretation. Even when down payment savings are assigned a higher beta weight than long-term savings, Alternative 2 does not appear to improve the path of H_t sufficiently to make it preferable. This indicates that the difference between the formulations is not only a matter of parameter values, but also of how the model structure translates preferences into actual saving allocations. This distinction is important in the context of the thesis. The down payment represents a goal-specific threshold that must be reached in order to enter the housing market. A utility formulation that shifts more wealth toward L_t may therefore appear attractive from a pure long-term wealth accumulation perspective, but will do so by delaying the housing goal. For the target group considered here, such a delay is not necessarily desirable.



(a) Alternative 1



(b) Alternative 2

Figure 5.3: Comparison of utility functions when $\beta = (0.2, 0.5, 0.15, 0.15)$

Figure 5.3 highlights the difference in spending behavior. The figure shows that the first utility formulation reaches a cumulative spending of around 600 KSEK over the 15 years, whereas the second alternative reaches around 800 KSEK. Therefore, the formulations also create different spending behavior. Since the model is normative rather than descriptive, higher spending is not considered preferable simply because

it may better resemble actual behavior. In a banking advisory context, lower spending can be preferable as long as it remains realistic, since it leaves more resources available for saving and can support earlier achievement of the housing goal.

Taken together, the results indicate that the difference between the two utility formulations is how strongly the model prioritizes the housing goal. This comparison is affected by the fact that Alternative 1 includes a housing bonus when the down payment target is reached, while Alternative 2 does not. To separate the effect of the target-relative formulation from the effect of the housing bonus, we therefore also test Alternative 2 with the same housing bonus.

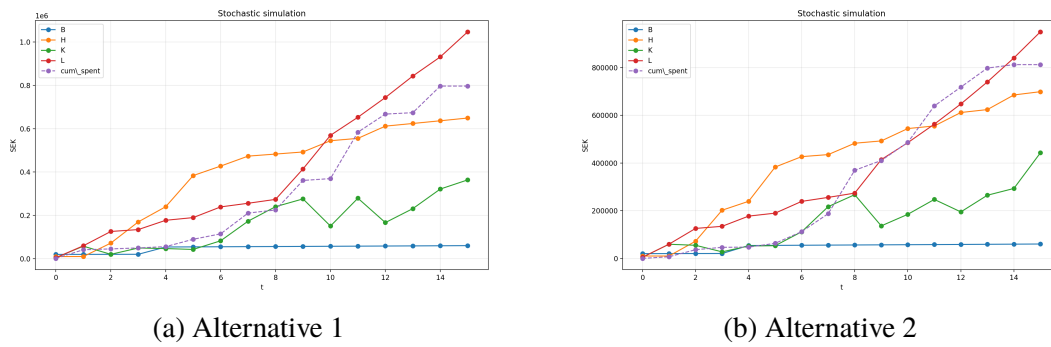


Figure 5.4: Comparison of utility functions when both include a housing bonus

Figure 5.4 shows that when the bonus is included, the simulated allocation paths become more similar to those generated by Alternative 1. Note that the graphs have different scales on the y-axis. This suggests that the difference is not because of the absolute versus target-relative formulation, but mainly because of the discrete incentive created by the housing bonus.

Based on this comparison, we proceed with Alternative 1 in the remaining analysis. This specification is simpler and easier to interpret, since the housing goal is represented by an absolute savings level together with a discrete bonus when the threshold is reached. In comparison to Alternative 2, adding the same bonus means that the housing target affects the utility function both through the target-relative logarithmic term and through the threshold bonus, which partly duplicates its influence. Alternative 1 is therefore preferred as the main specification, as it gives sufficient priority to the down payment goal while keeping the utility formulation more transparent. Moreover, Alternative 1 does not require explicit target levels for short-term and long-term savings. This makes the specification less dependent on assumptions about what constitutes an appropriate savings target for these components, and therefore more broadly applicable for individuals with different preferences and financial circumstances.

5.3 Comparison of Beta Specifications

Given the chosen utility function, we next examine which beta specifications generate the most intuitive saving behavior. As a starting point, we compare nine different beta specifications for an individual with an annual net income of 300 KSEK (25 KSEK per month) and a housing target of 1.5 MSEK, corresponding to an initial required down payment of 225 KSEK. The simulations are conducted over a ten-year horizon in order to reduce computational time, since several model specifications and parameter combinations are evaluated. In the first set of simulations with different beta specifications, no housing-goal bonus is included. This allows the effect of the beta weights to be isolated and makes it possible to assess whether the weights alone can generate intuitive allocation behavior. The beta specifications are designed to reflect different priorities across the savings categories. Specifications 1, 7, and 8 place a relatively high focus on housing savings, while 5 and 6 put more weight on buffer savings, and 9 gives more weight to long-term savings. Specification 2 serves as a neutral baseline with equal weights across all goals, while 3 and 4 increase the emphasis on buffer and housing savings while keeping short- and long-term savings equally weighted. For exact beta specifications, see Appendix A.1.

The evaluation is based primarily on the ability to reach the housing goal within the ten-year horizon, measured by the Monte Carlo success rates and average first hitting times for reaching the goal, while also evaluating how well each specification maintains a reasonable balance across the other savings components. The results in Appendix A.1 show that an increased focus on buffer savings does not provide any noticeable benefit, which leads to the exclusion of specifications 5 and 6. The baseline specification 2, as well as the intermediate specifications 3 and 4, are also discarded, since they do not perform well on any of the main dimensions considered.

A common feature of the top-performing beta specifications is a relatively high weight on housing savings, which supports faster accumulation toward the down payment while still preserving meaningful allocations to buffer and long-term savings. This makes them more attractive from a broader financial well-being perspective than specifications that either distribute savings too evenly and therefore do not meet the target goals, or place too much emphasis on long-term accumulation.

This trade-off can be illustrated by comparing Run 1 and Run 9. Run 9 generates a higher mean final long-term balance, approximately 687 KSEK compared with 581 KSEK in Run 1, but reaches the housing goal almost two years later on average (Appendix A.1). Since entering the housing market within a reasonable time horizon is a central objective of the model, this additional long-term accumulation is not sufficient to offset the weaker housing performance. For that reason, Run 1 is preferred to Run 9.

The strongest specifications are therefore Runs 1, 7, and 8, all of which combine high or near-complete success rates with the lowest average first hitting times, at approximately 5.1 years. These runs correspond to the beta vectors

$$(0.30, 0.40, 0.10, 0.20), \quad (0.25, 0.40, 0.15, 0.20), \quad (0.20, 0.50, 0.15, 0.15).$$

5.4 Evaluating the House Down Payment Bonus

Thereafter, the discrete bonus for reaching the housing goal is again included in the utility formulation. To evaluate whether it has a meaningful impact on the model outcomes, we compare the results obtained with and without the bonus term. The results show that the housing-goal bonus has a clear effect. For an individual earning 25 KSEK per month with a housing target of 1.5 MSEK, all beta specifications reach the housing goal faster when the bonus is included (Appendix A.2). The three strongest specifications remain Run 1, Run 7, and Run 8, but their average first hitting times decrease from around 5.1 years to around 4.15 years. The main interpretation is therefore that the bonus strengthens goal-oriented behavior around the down payment threshold. Without the bonus, the model may allocate substantial resources to housing savings and move close to the target, but still lack a strong enough incentive to cross the exact threshold. By adding a discrete reward once the target is reached, the model is encouraged to complete the housing goal rather than remain just below it. This effect is also visible for specifications that were previously less competitive. For instance, the equal-weight specification becomes substantially stronger, with the average first hitting time falling from 6.92 to 4.25 years.

This indicates that the bonus reinforces the intended goal-directed behavior of the model. The effect is visible not only in faster goal attainment, but also in the composition of savings. In general, the inclusion of the bonus shifts optimal allocations somewhat more toward housing savings and away, in particular, from long-term savings. At the same time, the model does not neglect the remaining components entirely: buffer holdings remain at adequate levels, and the better-performing specifications still preserve meaningful long-term accumulation.

Together, the comparison confirms that the bonus is an important part of the final model, and its value is further analyzed through a sensitivity analysis.

5.5 Assessing Different Scenarios

5.5.1 Performance when the housing target becomes more demanding

To test the robustness of the beta specifications, the housing target is then increased from 1.5 MSEK to 2.5 MSEK while keeping income fixed at 25 KSEK per month (Appendix A.3). This creates a more demanding scenario where the required down payment becomes substantially harder to achieve.

The results show that differences between beta specifications become much more pronounced. More specifically, only the profiles with a sufficiently strong emphasis on housing savings remain highly successful. Run 8, with the highest housing weight, becomes the best-performing specification regarding the down payment goal, having a success rate of 99.4 percent and the shortest average first hitting time of 6.69 years. Runs 1 and 7 also perform strongly, with success rates above 98 percent and only slightly longer hitting times.

By contrast, Run 2 with equal weights achieves a success rate of only 14.2 percent, while Run 9 with long-term savings focus drops to 4.3 percent. These results suggest that the housing goal will rarely be reached in runs where the beta for the goal is relatively low. This reinforces the idea that when the housing target becomes more demanding, the trade-off between competing savings goals becomes tighter. In such a setting, the model must give housing a sufficiently high priority for the goal to remain attainable, while still preserving reasonable outcomes in the buffer and long-term savings components.

Runs 1, 7, and 8 all perform adequately in this more demanding scenario, with high success rates and similar average first hitting times for the down payment. The main difference is the trade-off between short- and long-term savings. Runs 7 and 8 can be grouped together in that they place relatively more emphasis on short-term savings with a higher accumulated spending. In contrast, Run 1 achieves a similarly high success rate while preserving a higher level of long-term savings at the expense of lower short-term savings.

5.5.2 Higher income scenario

We next consider a higher income level of 35 KSEK per month together with the more demanding housing target of 2.5 MSEK. The main insight from this scenario is not whether the model can reach the housing goal, since this is already feasible for all specifications, but rather how the trade-off between first hitting time and the accumulated savings in each category changes when the budget constraint is relaxed.

Similar to the demanding housing scenario, Run 1 preserves a noticeably higher average long-term savings, approximately 819 KSEK, compared to around 756 KSEK for Run 7 and 637 KSEK for Run 8 (Appendix A.4). The difference is once more due to higher short-term spending in Runs 7 and 8. This suggests that in the higher-income case, Run 1 may be preferred over Runs 7 and 8 if the consumer wants to prioritize long-term savings over short-term spending.

5.5.3 Constrained income scenario

We also consider a constrained income level of 17 KSEK per month when the desired housing price is 1.5 MSEK. In this setting, Run 1 becomes the strongest specification, with a success rate of 89.5 percent and an average first hitting time of 7.20 years (Appendix A.7). By comparison, Run 8 has the second highest success rate of only 21.9 percent, despite placing the highest weight on housing savings. The key difference appears to be that Run 1 combines a strong housing focus with a more balanced allocation under a constrained budget, where the difference between income and expenses is small. This is reflected in the savings outcomes. Run 1 accumulates a higher average long-term balance of about 194 KSEK compared to Run 8, which reaches roughly 180 KSEK, while still achieving much higher success in reaching the housing goal.

A likely explanation is that Run 1 produces lower cumulative spending, which makes the housing goal more attainable under limited financial margins. Run 1 has approximately 159 KSEK in accumulated spending whereas Run 8 has approximately 183 KSEK. The level of short-term savings is also lower in Run 1. In other words, the specification captures that when income is constrained, reaching the down payment target requires a more restrictive and savings-oriented allocation pattern. Simply assigning a high weight to housing savings is therefore not sufficient if the resulting policy still allows too much spending or fails to preserve enough resources over time.

The comparison suggests that in a constrained income setting, the choice of beta specification does not merely result in a marginal difference in behavior as we saw in the other scenarios. Run 1 substantially outperforms the other alternatives, while still maintaining higher long-term savings. This indicates that a more balanced but disciplined allocation strategy is significantly more robust under tight budget conditions, whereas specifications that place excessive emphasis on housing savings do not necessarily translate this priority into effective goal attainment.

5.5.4 Constrained income combined with a demanding housing target

When the constrained income scenario is combined with the more demanding housing target of 2.5 MSEK, almost none of the specifications reach the down payment goal. This result is expected, since the required down payment becomes too large relative to the individual's saving capacity. In this case, the model often allocates little or no resources to housing savings, not because housing is unimportant, but because the target is perceived as practically unattainable within the planning horizon. This effect can be observed by studying the differences in housing savings in Appendix A.7 in comparison to Appendix A.8, where the inability to reach the target leads to a shift in allocations away from the down payment component.

5.6 Sensitivity Analysis

A sensitivity analysis of the housing down payment bonus was then performed. It reveals a clear non-linear effect. Initially, we evaluated outcomes when net income is 25 KSEK per month for the three best-performing beta specifications. At low bonus levels (around $0 < b \leq 0.4$), the impact on both behavior and outcomes is limited (Appendix A.9 and A.10). Around $b = 0.5$, the model begins to approach a threshold, where the model starts to shift toward more goal-oriented behavior, but has not yet fully transitioned. Outcomes improve significantly compared to no bonus, and success rates reach relatively high levels. However, behavior is still relatively balanced, with a substantial share of savings allocated to long-term accumulation and only moderate emphasis on housing. When the bonus is increased to $b = 0.6$, the model exhibits a strong behavioral shift, with faster goal attainment and a higher allocation toward housing. For higher bonus levels of $b = 0.7$ and $b = 0.8$, it continues to improve housing goal attainment, but with diminishing marginal returns.

The threshold effect is even more evident in the constrained income scenario (17 KSEK per month). At a bonus of 0.4, success rates remain low, and the model struggles to accumulate sufficient housing savings. At $b = 0.5$, outcomes improve, but performance remains uneven across specifications as previously discussed. In contrast, increasing the bonus to $b = 0.6$ induces a clear behavior shift where success rates increase significantly. For example, Run 8 dramatically improves the success rate from 21.9 percent to 86.2 percent. This shows that while a bonus at 0.5 is close to the critical level, 0.6 more reliably enforces goal-oriented behavior under tight budget constraints. However, this budget-constrained scenario is intended to model cases in which the goals are hard to achieve and $b = 0.5$ maintains a more balanced allocation of resources, which is preferable and represents a more robust model behavior.

In contrast, in high-income scenarios (35 KSEK/month), the threshold effect largely disappears. Success rates are already 100 percent even at lower bonus levels, and increasing the bonus mainly affects the speed of goal attainment and the allocation margin between the housing and long-term savings goals.

Overall, the results show that the bonus has a non-linear effect on the allocations. With a high enough bonus, resources shift decisively toward housing goals, primarily at the expense of long-term savings and consumption, while buffer stability is largely preserved. By comparing bonus levels 0.5 and 0.6, it is preferable to keep 0.5 as it leads to more robust and balanced outcomes, particularly in constrained environments.

5.7 Comparison of the Strongest Candidates

Across all tested scenarios, Run 1, Run 7, and Run 8 stand out as the most relevant candidates.

Run 1 (0.30,0.40,0.10,0.20) appears to be the most robust specification overall. It performs strongly in all feasible scenarios and is especially the best choice in the most constrained simulation, implying that this might be the best fit in the event of high fixed costs in relation to income (Appendix A.6). Notably, in this simulation the added housing bonus utility is crucial to make the target attainable within the ten-period time frame. This indicates that its combination of relatively high housing weight, moderate buffer weight, low short-term weight, and a moderately high long-term weight provides a good balance between goal pursuit and financial stability.

Run 7 (0.25,0.40,0.15,0.20) can be interpreted as a compromise specification. It consistently performs well and remains close to the top performer in most scenarios, but it is rarely the single best case. Its advantage lies in stability and robustness rather than delivering optimal results under ideal conditions.

Run 8 (0.20,0.50,0.15,0.15) is frequently the fastest in attaining the housing down payment goal and particularly strong when the housing target is demanding but still realistically attainable. At the same time, it becomes less attractive in highly budget-constrained and/or low-income scenarios, where a more balanced preference structure appears to be beneficial.

Taken together, these results suggest a trade-off between speed and robustness. Run 8 is often the most efficient at reaching the housing threshold, while Run 1 is the most robust across very different economic conditions. Run 7 occupies an intermediate position and may therefore serve as a useful compromise case.

Chapter 6

Discussion

This chapter discusses the main findings, their implications for financial advisory, practical implementation, validity, and reliability.

6.1 Interpretation of the Main Results

The results show that the model captures meaningful trade-offs between competing financial goals under uncertainty. In particular, there is a clear tension between allocating resources toward the housing down payment and maintaining financial resilience through buffer savings and diversified long-term accumulation.

The comparison between the two utility alternatives and the introduction of the housing bonus highlights that the formulation of financial well-being plays a decisive role in shaping allocation behavior. The bonus should therefore be interpreted as an advisory design parameter rather than a directly observable preference parameter. The structure of the objective function and the beta specification determine how trade-offs are resolved, particularly when goals differ in nature, such as threshold-based targets (housing) versus continuous accumulation (long-term savings). The results therefore indicate that the relative prioritization of the goals should be defined in advance.

6.2 Implications for Financial Advisory

From a financial advisory perspective, the model illustrates how different prioritizations of financial goals may lead to different outcomes over time. Rather than prescribing a single optimal allocation, the framework provides a structured way to analyze the consequences of alternative strategies under uncertainty.

As stated, the results also show that the utility function plays a central role in determining how financial goals are prioritized. In an advisory context, this can be used as an advantage. Instead of relying only on static rules of thumb, advisors could use the model to translate advisory principles into a dynamic framework. For example, a bank may want to prioritize sufficient liquidity before housing savings, or give stronger weight to housing progress for customers who are close to reaching a down payment target. The model can then be used to simulate how such priorities affect allocations over time under different income and risk conditions.

This is relevant because advisory decisions often involve trade-offs that are difficult to evaluate separately. For example, accelerating progress toward a housing goal may come at the cost of reduced financial resilience, while maintaining a stronger buffer may delay housing entry but improve stability under adverse scenarios. By making these trade-offs explicit, the model could support more transparent and individualized advisory, provided that its outputs are validated and communicated properly.

6.3 Implementation in Financial Advisory

From an implementation perspective, the model could be used in several ways within a banking advisory context, depending on how much of the advisory process is intended to be automated.

One possible application is a customer-facing digital tool. For example, it could be integrated into a website or mobile banking platform. In this setting, customers could enter their financial situation and goals, and receive personalized allocation strategies. This would make the framework highly scalable and easy to access. However, such an implementation would also require a high level of trust in the model outputs. The model outputs would need to be stable, interpretable, and communicated in a way that users can understand. If the output is unclear or too sensitive to small changes in input values, there is a risk that customers either misinterpret the recommendation or lose trust not only in the tool, but also in the financial institution.

A more realistic short-term implementation may therefore be to use the model as a support tool in financial advisory. In this setting, the model would not replace the advisor, but instead provide structured model-based support during the advisory process. The advisor could use the model to compare different allocation strategies, test how sensitive the outcome is to changes in assumptions, and explain trade-offs between goals. This could make it possible to provide more detailed and individualized advice than is typically feasible within the limited time available in a standard advisory meeting today. Such an implementation would combine scalability with human interpretation. The model can process individual financial

information and produce structured decision support, while the advisor remains responsible for explaining the results, assessing whether they are reasonable, and adapting them to the customer's broader situation. In this way, the framework can support more personalized advisory without requiring a fully automated decision process.

6.4 Validity

The validity of the present study depends on whether the model captures a meaningful representation of financial well-being in the intended advisory context. A strength of the framework is that it translates concrete and financially relevant goals into a dynamic optimization problem. This supports the model's relevance for exploring banking advisory applications. The calibration to Swedish macroeconomic data improves the contextual grounding of the model, although it does not by itself ensure external validity. Furthermore, we have only modeled two of many possible life events that can occur, meaning that the negative impact of, for example, not maintaining a sufficient buffer might be substantially higher in real life.

While reliable estimates for housing market growth and asset returns were obtained from available data sources, these assumptions were also discussed and validated in collaboration with supervisors at Handelsbanken. This helps to ensure that the parameter choices are aligned with practical with a bank's internal macroeconomic outlook. Consequently, the model is grounded in empirically informed and practically relevant assumptions regarding future economic conditions. Note that the macroeconomic assumptions are stationary, such as having a constant inflation rate, which limits their ability to capture cyclical fluctuations and changing economic environments. Instead of explicitly modeling macroeconomic events, systemic shocks are incorporated through their effects on individual household finances. For example, a financial crisis is one of the exogenous shocks modeled and is calibrated using empirical evidence from [16]. The study is based on Danish data from 2007–2008, and may therefore not be fully transferable to the current Swedish context. However, given the institutional and macroeconomic similarities between the two countries, the estimates are considered a reasonable approximation for modeling purposes, while still representing a source of uncertainty in the analysis.

Regarding the structure of the model, it also reflects fundamental economic constraints faced by single households, such as intertemporal budget constraints and precautionary saving motives. As such, the model can produce economically intuitive results, such as increased allocation toward housing down payment when the housing goal becomes more demanding as well as more balanced allocations under tighter budget constraints.

Nevertheless, financial well-being is a broader concept than what is captured in this thesis. In reality, it includes subjective dimensions such as financial stress, perceived control, and personal satisfaction. Thus, the model reflects a deliberately delimited and practical definition of the concept based on measurable financial goals. It should be interpreted as capturing four quantitative dimensions of financial well-being rather than the larger and more abstract concept.

Additionally, an important limitation of the model arises in scenarios where the housing down payment target is not attainable within the time frame. In such cases, the bonus utility associated with reaching the housing goal is never activated, which alters the incentives for the utility embedded in the optimization model. A reasonable policy should encourage gradual progress toward the target rather than abandoning the goal entirely. Here, the model may exhibit distorted behavior and will instead prioritize alternative savings categories and thereby consumption, not because they are inherently preferred, but because the housing goal no longer provides marginal utility gains. This highlights a structural limitation of threshold-based utility components, as they rely on the feasibility of the target. It also suggests that the model's recommendations should be interpreted with caution in highly constrained scenarios, where certain financial goals may be unrealistic to achieve within the given time horizon. In such cases, an appropriate interpretation of the results is that the housing goal is unattainable given the individual's income, fixed costs, and planning horizon. From a banking advisory perspective, this implies that recommendations should instead focus on setting a more reasonable housing target, extending the time horizon, or even reducing fixed expenses.

6.5 Reliability

Reliability is supported by the fact that the implementation of the model is transparent and reproducible. The numerical solution procedure is deterministic, implying that identical parameter inputs yield identical policy functions. The structure of the dynamic programming model, the state transitions, and the Monte Carlo simulation are explicitly defined, which strengthens consistency and replicability.

Reliability is further supported by evaluating robustness across a wide range of parameter configurations and economic scenarios, including varying income levels and housing targets. In addition, the use of Monte Carlo simulations enables the model to be evaluated across a large number of possible future scenarios, where key variables such as income, spending, and life events evolve stochastically. Rather than relying on a single simulated path, the model is assessed based on the distribution of outcomes across many realizations. This reduces the risk that the results are driven by specific or unusually favorable realizations of uncertainty and instead provides

a broader assessment of policy performance under modeled uncertainty. At the same time, the results depend on the chosen resolution of the state and action grids. A finer grid can capture more detailed allocation behavior, but also increases the computational burden substantially. The grid choices are therefore partly a modeling judgment, where the level of detail must be balanced against available computational capacity and feasible run time. In this thesis, the chosen discretization is a practical compromise rather than a uniquely optimal grid structure.

Chapter 7

Conclusions and Future Work

This chapter summarizes the thesis, and presents conclusions and suggestions for future work.

7.1 Conclusions

This thesis set out to examine how a composite index of financial well-being can be constructed and integrated into a dynamic optimization framework that produces reasonable allocation behavior under exogenous risks. The results show that financial well-being can be operationalized through a set of measurable financial components that are relevant in an individual banking advisory context, namely buffer savings, housing down payment savings, short-term savings, and long-term savings. By representing these components as state variables and aggregating them in a utility function, the thesis demonstrates how financial well-being can be incorporated into a dynamic programming model.

The main conclusion is that dynamic programming can serve as a suitable and flexible framework for analyzing the allocation problem considered in this thesis. The model is able to represent a central trade-off between goal attainment and financial stability. In particular, it indicates that stronger emphasis on housing savings leads to faster achievement of the down payment goal, while more balanced specifications perform better when income is constrained or when financial margins are limited. This suggests that, with the scenarios considered, the preferred allocation rule depends on the individual's financial situation and that a single static rule is unlikely to be suitable across all scenarios.

The results also show that the formulation of the utility function is important. The housing bonus was a particularly important addition because it creates an incentive to fully reach the down payment threshold rather than remain close to it. However,

the sensitivity analysis also indicates that the bonus must be chosen carefully, since a too strong emphasis on the housing goal can reduce the balance between the savings components.

Overall, the thesis demonstrates how financial well-being can be translated into measurable financial goals and integrated into a dynamic programming framework. Under the selected assumptions, the resulting model produces financially interpretable allocation behavior under uncertainty and can be used to analyze trade-offs between liquidity, housing savings, short-term consumption needs, and long-term financial development. While the model relies on simplifying assumptions and should not be interpreted as a complete representation of individual financial well-being, it provides an initial modeling foundation for how dynamic programming can be applied in an individual banking advisory context. Rather than being a directly implementable advisory tool, the model should be seen as a baseline framework that can be further developed, refined, and empirically validated in future work.

7.2 Future Work

Several extensions could strengthen the framework and make it more realistic in future implementations. First, the model could be extended to include the individual's housing situation as an additional component of the utility function. In the current model, the housing goal is represented by the accumulation of a down payment, while the financial situation after a potential home purchase is not modeled explicitly. A future extension could distinguish between renting and owning a residence, where ownership is associated with both a change in utility and a different set of financial constraints. After a home purchase, rental costs would be replaced by mortgage debt, amortization requirements, and other housing-related costs. This would provide a more realistic representation of the financial consequences of entering the housing market, while also capturing the potential increase in well-being associated with owning a home.

Second, the model could be extended to include two-person households. While the individual approach is suitable for the younger target group considered in this thesis, financial decisions are often made jointly in later life stages. A household-level version of the model could include shared expenses, multiple incomes, and joint saving goals. It could also allow for transitions between one-person and two-person households, for example when individuals move in with a partner or separate. However, such an extension would require additional state variables and transition probabilities, which would increase both the empirical requirements and the computational complexity of the model.

Third, the treatment of short-term savings could be made more realistic. In the

current model, withdrawals from short-term savings are determined by stochastic spending shares, which captures that these savings are intended to be used, but may also generate unrealistically large withdrawals, especially in later stages of the model where the accumulated savings are higher. A future extension could introduce a more structured representation of planned expenditures, for example by keeping track of when short-term savings are accumulated and when the corresponding expenditure is expected to occur. This could make short-term saving behavior more gradual and realistic over time. However, it would also increase the complexity of the state space and the computational time required to solve the model.

Finally, future versions of the model could explore broader interpretations of financial well-being. The present thesis focuses on measurable financial goals, but financial well-being may also include subjective dimensions such as perceived financial security, financial stress, and individual consumption preferences. Incorporating such dimensions would require additional qualitative or survey-based data on how individuals value different financial outcomes. This could make the model better able to reflect heterogeneous preferences, since the same allocation may be perceived as secure by one individual but restrictive or stressful by another. However, it would also make the framework more difficult to generalize and validate.

Overall, these extensions would make the framework more realistic and applicable to a wider range of financial situations. At the same time, they would also increase the dimensionality of the state space and the amount of data required for calibration. Future work should therefore balance added realism against computational tractability and practical usefulness in a banking advisory context.

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Appendix A

Simulation Results

A.1 Baseline scenario without housing bonus

$I = 25$ KSEK/month, $P = 1.5$ MSEK, $b = 0$

Run	Beta (B,H,K,L)	Avg first hitting time (t)	Avg mean(B)	Avg mean(H)	Avg mean(K)	Avg mean(L)	Avg mean cumulative spending	MC success rate (%)	First hitting time distribution	Avg policy shares
1	(0.30,0.40,0.10,0.20)	5.10	58820	371603	229228	581343	395996	100.0	(0,0,0,0,0,4651,184,157,8,0,0)	(3.4,24.9,40.9,30.9)
2	(0.25,0.25,0.25,0.25)	6.92	57709	326478	245164	505231	503875	98.7	(0,0,0,0,0,847,3657,423,6,0)	(3.1,21.4,49.4,26.1)
3	(0.30,0.30,0.20,0.20)	5.31	56151	318827	261516	419599	566929	100.0	(0,0,0,0,0,3721,999,270,10,0,0)	(3.4,21.1,53.7,21.7)
4	(0.35,0.35,0.15,0.15)	5.22	58742	320085	262614	410072	569874	99.8	(0,0,0,0,0,4145,599,237,9,0,0)	(3.4,21.5,53.9,21.2)
5	(0.40,0.30,0.15,0.15)	5.36	57631	320392	259762	416028	568114	99.9	(0,0,0,0,0,3726,738,521,9,0,0)	(3.4,21.3,53.7,21.6)
6	(0.45,0.25,0.15,0.15)	5.28	57508	317383	261625	414044	572135	100.0	(0,0,0,0,0,3893,807,288,11,1,0)	(3.5,21.0,54.1,21.4)
7	(0.25,0.40,0.15,0.20)	5.11	57937	323175	241046	476706	526328	100.0	(0,0,0,0,0,4574,320,99,6,0,0)	(3.2,21.6,49.9,25.3)
8	(0.20,0.50,0.15,0.15)	5.07	58314	375037	254903	442871	490633	100.0	(0,0,0,0,0,4723,227,50,0,0,0)	(3.2,25.1,48.3,23.3)
9	(0.20,0.25,0.15,0.40)	6.88	56657	344468	219298	687079	368612	99.8	(0,0,0,0,0,1018,3572,385,13,0)	(3.0,22.8,38.5,35.6)

A.2 Baseline scenario with housing bonus

$I = 25$ KSEK/month, $P = 1.5$ MSEK, $b = 0.5$

Run	Beta (B,H,K,L)	Avg first hitting time (t)	Avg mean(B)	Avg mean(H)	Avg mean(K)	Avg mean(L)	Avg mean cumulative spending	MC success rate (%)	First hitting time distribution	Avg policy shares
1	(0.30,0.40,0.10,0.20)	4.12	62445	374301	230992	559382	403685	100.0	(0,0,0,0,4505,374,121,0,0,0,0)	(3.5,25.2,41.4,29.9)
2	(0.25,0.25,0.25,0.25)	4.25	58209	344307	264958	399733	548476	100.0	(0,0,0,0,4033,717,241,9,0,0,0)	(3.2,23.2,52.6,21.1)
3	(0.30,0.30,0.20,0.20)	4.28	60613	346548	263544	396366	546678	100.0	(0,0,0,0,3796,1020,184,0,0,0,0)	(3.4,23.4,52.4,20.8)
4	(0.35,0.35,0.15,0.15)	4.42	58102	376075	262493	400543	516166	100.0	(0,0,0,0,3063,1770,167,0,0,0,0)	(3.2,25.4,50.2,21.2)
5	(0.40,0.30,0.15,0.15)	4.73	58104	375816	256859	421092	502224	100.0	(0,0,0,0,1508,3325,167,0,0,0,0)	(3.3,25.4,49.1,22.3)
6	(0.45,0.25,0.15,0.15)	4.44	58644	351364	260967	405153	535604	100.0	(0,0,0,0,3042,1728,221,8,0,0,0)	(3.4,23.6,51.5,21.5)
7	(0.25,0.40,0.15,0.20)	4.19	60391	375919	239252	445782	496633	100.0	(0,0,0,0,4181,682,137,0,0,0,0)	(3.4,25.4,47.6,23.7)
8	(0.20,0.50,0.15,0.15)	4.19	60440	375909	264651	394087	520609	100.0	(0,0,0,0,4179,687,133,1,0,0,0)	(3.3,25.4,50.6,20.7)
9	(0.20,0.25,0.15,0.40)	5.27	53293	324361	226768	682058	388587	100.0	(0,0,0,0,2,3759,1124,107,8,0,0)	(3.2,21.5,40.0,35.4)

A.3 Higher income and demanding housing target without housing bonus

$I = 35$ KSEK/month, $P = 2.5$ MSEK, $b = 0$

Run	Beta (B,H,K,L)	Avg first hitting time (t)	Avg mean(B)	Avg mean(H)	Avg mean(K)	Avg mean(L)	Avg mean cumulative spending	MC success rate (%)	First hitting time distribution	Avg policy shares
1	(0.30,0.40,0.10,0.20)	7.49	59360	600378	184692	418279	356588	99.3	(0,0,0,0,232,244,1898,2135,381,73)	(3.4,39.6,35.1,21.9)
2	(0.25,0.25,0.25,0.25)	9.00	57366	449726	243246	369788	496755	14.2	(0,0,0,0,0,0,0,0,711,0)	(3.1,29.3,48.8,18.8)
3	(0.30,0.30,0.20,0.20)	7.91	58093	531021	238214	348241	432449	79.1	(0,0,0,0,0,0,1093,2181,643,40)	(3.2,35.2,43.6,18.0)
4	(0.35,0.35,0.15,0.15)	7.69	56538	578706	209415	347830	414367	95.0	(0,0,0,0,0,0,2955,666,780,351)	(3.1,38.2,40.6,18.1)
5	(0.40,0.30,0.15,0.15)	7.69	56330	564957	220313	346919	416463	91.8	(0,0,0,0,0,0,2799,685,857,250)	(3.1,37.3,41.5,18.0)
6	(0.45,0.25,0.15,0.15)	8.01	56838	534857	231946	350800	432126	79.4	(0,0,0,0,0,0,1094,1870,861,146)	(3.3,35.2,43.2,18.3)
7	(0.25,0.40,0.15,0.20)	7.66	57818	592226	198827	349024	410714	98.0	(0,0,0,0,0,0,2910,910,888,190)	(3.2,39.0,39.7,18.1)
8	(0.20,0.50,0.15,0.15)	6.69	58652	630150	208929	324230	382226	99.4	(0,0,0,0,0,0,2835,1168,695,236,36)	(3.2,41.6,38.4,16.8)
9	(0.20,0.25,0.15,0.40)	9.42	51025	451453	218040	549466	384386	4.3	(0,0,0,0,0,0,0,0,124,89)	(3.0,29.4,39.4,28.2)

A.4 Higher income and demanding housing target with housing bonus

$I = 35$ KSEK/month, $P = 2.5$ MSEK, $b = 0.5$

Run	Beta (B,H,K,L)	Avg first hitting time (t)	Avg mean(B)	Avg mean(H)	Avg mean(K)	Avg mean(L)	Avg mean cumulative spending	MC success rate (%)	First hitting time distribution	Avg policy shares
1	(0.30,0.40,0.10,0.20)	3.18	82554	896112	224511	818795	523353	100.0	(0,0,0,4263,630,33,72,2,0,0,0)	(3,2,37,7,31,7,27,5)
2	(0.25,0.25,0.25,0.25)	4.96	80394	617926	331311	869678	664875	96.2	(0,0,0,0,543,4067,62,136,1,0,0)	(3,0,26,2,41,8,29,1)
3	(0.30,0.30,0.20,0.20)	5.10	79609	752669	305475	766056	650957	99.9	(0,0,0,0,500,3966,219,177,112,12,10)	(3,0,31,6,40,2,25,2)
4	(0.35,0.35,0.15,0.15)	4.11	77219	873651	293827	627666	645179	100.0	(0,0,0,0,4678,90,217,13,2,0,0)	(3,0,36,6,39,5,20,9)
5	(0.40,0.30,0.15,0.15)	4.12	77229	867984	294483	630675	643183	100.0	(0,0,0,0,4682,86,212,14,6,0,0)	(3,0,36,4,39,5,21,0)
6	(0.45,0.25,0.15,0.15)	4.15	77203	831982	297734	667177	644577	100.0	(0,0,0,0,4600,104,274,15,6,1,0)	(3,0,35,0,39,7,22,4)
7	(0.25,0.40,0.15,0.20)	3.29	82291	871349	273319	755763	563105	100.0	(0,0,0,3817,998,97,78,9,0,1,0)	(3,2,36,6,35,1,25,1)
8	(0.20,0.50,0.15,0.15)	3.30	81039	898599	293559	637304	620409	100.0	(0,0,0,3786,1043,89,73,7,2,0,0)	(3,1,37,7,38,2,21,0)
9	(0.20,0.25,0.15,0.40)	5.10	79899	630871	246167	1059487	582236	94.6	(0,0,0,0,4501,59,128,14,28,0)	(3,0,26,8,34,9,35,3)

A.5 Higher income and highly demanding housing target without housing bonus

$I = 35$ KSEK/month, $P = 3.5$ MSEK, $b = 0$

Run	Beta (B,H,K,L)	Avg first hitting time (t)	Avg mean(B)	Avg mean(H)	Avg mean(K)	Avg mean(L)	Avg mean cumulative spending	MC success rate (%)	First hitting time distribution	Avg policy shares
1	(0.30,0.40,0.10,0.20)	5.39	81356	770967	225083	925387	572524	100.0	(0,0,0,0,0,3855,656,279,131,72,7)	(3,1,32,1,33,9,30,9)
2	(0.25,0.25,0.25,0.25)	5.65	80687	502662	333145	945812	723619	52.5	(0,0,0,0,0,1732,304,422,98,69,0)	(3,1,21,2,44,4,31,3)
3	(0.30,0.30,0.20,0.20)	5.50	80147	674492	307359	837294	673436	99.5	(0,0,0,0,0,3843,367,274,423,24,46)	(3,1,28,1,41,2,27,7)
4	(0.35,0.35,0.15,0.15)	5.44	77188	763345	296806	776599	649088	100.0	(0,0,0,0,0,3789,533,386,265,26,1)	(3,0,31,7,39,7,25,6)
5	(0.40,0.30,0.15,0.15)	5.43	77199	724118	297924	812539	648737	100.0	(0,0,0,0,0,3811,537,366,265,12,7)	(3,0,30,2,39,9,27,0)
6	(0.45,0.25,0.15,0.15)	5.49	77177	702204	299756	825472	658726	99.8	(0,0,0,0,0,3792,453,373,306,24,42)	(3,0,29,2,40,3,27,5)
7	(0.25,0.40,0.15,0.20)	5.36	80857	736971	274858	861309	620938	100.0	(0,0,0,0,0,3907,588,346,139,15,5)	(3,0,30,8,37,7,28,5)
8	(0.20,0.50,0.15,0.15)	4.31	80258	827633	294973	665304	667022	100.0	(0,0,0,0,4083,421,375,115,6,0,0)	(3,0,34,6,40,4,22,0)
9	(0.20,0.25,0.15,0.40)	6.58	79768	447203	245769	1259525	615606	37.2	(0,0,0,0,0,1353,31,379,87,8)	(3,0,19,0,36,2,41,8)

A.6 Constrained income scenario without housing bonus

$I = 17$ KSEK/month, $P = 1.5$ MSEK, $b = 0$

Run	Beta (B,H,K,L)	Avg first hitting time (t)	Avg mean(B)	Avg mean(H)	Avg mean(K)	Avg mean(L)	Avg mean cumulative spending	MC success rate (%)	First hitting time distribution	Avg policy shares
1	(0.30,0.40,0.10,0.20)	10.00	40937	277592	63794	233881	187817	0.1	(0,0,0,0,0,0,0,0,0,3)	(4.0,37.2,33.6,25.2)
2	(0.25,0.25,0.25,0.25)	–	36846	190639	125934	279245	199887	0.0	(0,0,0,0,0,0,0,0,0,0)	(3.4,24.6,43.1,28.9)
3	(0.30,0.30,0.20,0.20)	–	38381	227352	113716	225369	204414	0.0	(0,0,0,0,0,0,0,0,0,0)	(3.6,30.3,42.4,23.7)
4	(0.35,0.35,0.15,0.15)	–	40000	274938	98711	192807	196649	0.0	(0,0,0,0,0,0,0,0,0,0)	(3.9,36.7,39.2,20.1)
5	(0.40,0.30,0.15,0.15)	–	40951	260502	104010	200688	197395	0.0	(0,0,0,0,0,0,0,0,0,0)	(4.1,34.8,40.1,21.0)
6	(0.45,0.25,0.15,0.15)	–	41274	232463	113178	220579	201619	0.0	(0,0,0,0,0,0,0,0,0,0)	(4.1,30.8,42.0,23.1)
7	(0.25,0.40,0.15,0.20)	9.00	38108	266577	93880	213980	193263	0.0	(0,0,0,0,0,0,0,0,1,0)	(3.6,35.6,38.2,22.6)
8	(0.20,0.50,0.15,0.15)	9.01	37438	283327	103195	186989	191270	2.6	(0,0,0,0,0,0,0,0,127,1)	(3.4,38.0,39.1,19.5)
9	(0.20,0.25,0.15,0.40)	–	38434	196069	93096	327110	184999	0.0	(0,0,0,0,0,0,0,0,0,0)	(3.5,25.3,36.9,34.4)

A.7 Constrained income scenario with housing bonus

$I = 17$ KSEK/month, $P = 1.5$ MSEK, $b = 0.5$

Run	Beta (B,H,K,L)	Avg first hitting time (t)	Avg mean(B)	Avg mean(H)	Avg mean(K)	Avg mean(L)	Avg mean cumulative spending	MC success rate (%)	First hitting time distribution	Avg policy shares
1	(0.30,0.40,0.10,0.20)	7.20	41219	308036	99006	193735	158594	89.5	(0,0,0,0,0,0,0,3932,192,342,8)	(4.0,41.6,33.7,20.7)
2	(0.25,0.25,0.25,0.25)	8.96	36900	290722	106516	182813	186577	15.9	(0,0,0,0,0,0,0,28,767,0)	(3.4,38.6,39.0,19.0)
3	(0.30,0.30,0.20,0.20)	8.98	37933	291120	104313	180076	187161	14.5	(0,0,0,0,0,0,0,4,6,715,0)	(3.5,38.9,38.8,18.8)
4	(0.35,0.35,0.15,0.15)	8.99	40071	292528	102964	179239	185517	15.2	(0,0,0,0,0,0,0,3,5,753,0)	(3.9,39.2,38.3,18.7)
5	(0.40,0.30,0.15,0.15)	8.98	41169	292686	102962	177921	184474	16.3	(0,0,0,0,0,0,0,5,4,804,0)	(4.0,39.2,38.2,18.5)
6	(0.45,0.25,0.15,0.15)	8.99	41250	292339	103732	178314	184137	16.5	(0,0,0,0,0,0,0,0,9,817,0)	(4.0,39.1,38.3,18.6)
7	(0.25,0.40,0.15,0.20)	8.98	38202	292312	98281	187464	184722	15.2	(0,0,0,0,0,0,0,4,4,753,0)	(3.6,39.2,37.6,19.7)
8	(0.20,0.50,0.15,0.15)	8.61	37787	295172	105192	179567	183257	21.9	(0,0,0,0,0,0,0,101,228,766,0)	(3.4,39.9,38.1,18.7)
9	(0.20,0.25,0.15,0.40)	9.00	39058	274263	92479	276692	149094	3.0	(0,0,0,0,0,0,0,0,0,148,0)	(3.3,35.5,32.3,29.0)

A.8 Constrained income and demanding housing target with housing bonus

$I = 17$ KSEK/month, $P = 2.5$ MSEK, $b = 0.5$

Run	Beta (B,H,K,L)	Avg first hitting time (t)	Avg mean(B)	Avg mean(H)	Avg mean(K)	Avg mean(L)	Avg mean cumulative spending	MC success rate (%)	First hitting time distribution	Avg policy shares
1	(0.30,0.40,0.10,0.20)	–	40830	300724	70125	210980	180018	0.0	(0,0,0,0,0,0,0,0,0,0)	(4.0,40.1,33.5,22.4)
2	(0.25,0.25,0.25,0.25)	–	36846	190639	125934	279245	199887	0.0	(0,0,0,0,0,0,0,0,0,0)	(3.4,24.6,43.1,28.9)
3	(0.30,0.30,0.20,0.20)	–	38381	227352	113716	225369	204414	0.0	(0,0,0,0,0,0,0,0,0,0)	(3.6,30.3,42.4,23.7)
4	(0.35,0.35,0.15,0.15)	–	39754	298533	90024	185576	187747	0.0	(0,0,0,0,0,0,0,0,0,0)	(3.9,39.7,37.1,19.3)
5	(0.40,0.30,0.15,0.15)	–	40938	264631	102531	199512	195708	0.0	(0,0,0,0,0,0,0,0,0,0)	(4.1,35.3,39.8,20.9)
6	(0.45,0.25,0.15,0.15)	–	41288	239336	112720	211469	202399	0.0	(0,0,0,0,0,0,0,0,0,0)	(4.1,31.8,41.9,22.2)
7	(0.25,0.40,0.15,0.20)	–	38151	290386	86596	200207	187992	0.0	(0,0,0,0,0,0,0,0,0,0)	(3.6,38.6,36.7,21.1)
8	(0.20,0.50,0.15,0.15)	–	36846	321861	83440	178658	180266	0.0	(0,0,0,0,0,0,0,0,0,0)	(3.4,42.8,35.3,18.5)
9	(0.20,0.25,0.15,0.40)	–	38434	196069	93096	327110	184999	0.0	(0,0,0,0,0,0,0,0,0,0)	(3.5,25.3,36.9,34.4)

A.9 Demanding housing target with reduced housing bonus

$I = 25$ KSEK/month, $P = 2.5$ MSEK, $b = 0.4$

Run	Beta (B,H,K,L)	Avg first hitting time (t)	Avg mean(B)	Avg mean(H)	Avg mean(K)	Avg mean(L)	Avg mean cumulative spending	MC success rate (%)	First hitting time distribution	Avg policy shares
1	(0.30,0.40,0.10,0.20)	7.74	58203	593447	188718	410414	368947	99.1	(0,0,0,0,0,0,2078,2218,517,143)	(3.3,39.2,36.2,21.4)
2	(0.25,0.40,0.15,0.20)	7.73	57850	590357	200456	349977	409805	98.2	(0,0,0,0,0,0,2776,920,983,231)	(3.2,38.9,39.8,18.1)
3	(0.20,0.50,0.15,0.15)	7.44	58814	604715	209866	323748	405201	99.7	(0,0,0,0,0,0,3288,1270,354,71)	(3.2,39.9,40.0,16.8)

A.10 Demanding housing target with low housing bonus

$I = 25$ KSEK/month, $P = 2.5$ MSEK, $b = 0.3$

Run	Beta (B,H,K,L)	Avg first hitting time (t)	Avg mean(B)	Avg mean(H)	Avg mean(K)	Avg mean(L)	Avg mean cumulative spending	MC success rate (%)	First hitting time distribution	Avg policy shares
1	(0.30,0.40,0.10,0.20)	7.72	58407	595368	191941	377356	392445	98.9	(0,0,0,0,0,0,2702,1240,710,293)	(3.3,39.2,37.9,19.5)
2	(0.25,0.40,0.15,0.20)	7.78	58058	589495	199450	354791	407428	98.2	(0,0,0,0,0,0,2776,708,1152,272)	(3.2,38.7,39.6,18.5)
3	(0.20,0.50,0.15,0.15)	7.44	58814	604715	209866	323748	405201	99.7	(0,0,0,0,0,0,3288,1270,354,71)	(3.2,39.9,40.0,16.8)

A.11 Demanding housing target with increased housing bonus

$I = 25$ KSEK/month, $P = 2.5$ MSEK, $b = 0.6$

Run	Beta (B,H,K,L)	Avg first hitting time (t)	Avg mean(B)	Avg mean(H)	Avg mean(K)	Avg mean(L)	Avg mean cumulative spending	MC success rate (%)	First hitting time distribution	Avg policy shares
1	(0.30,0.40,0.10,0.20)	7.26	59428	609215	186031	412259	351607	99.3	(0,0,0,0,0,393,575,1773,1873,285,66)	(3.4,40.1,34.9,21.6)
2	(0.25,0.40,0.15,0.20)	7.40	59856	600284	202052	350775	397023	99.0	(0,0,0,0,0,228,3194,940,506,80)	(3.3,39.7,38.9,18.1)
3	(0.20,0.50,0.15,0.15)	6.48	59106	637905	208675	323504	373557	99.7	(0,0,0,0,0,24,3641,490,578,229,25)	(3.3,42.1,37.8,16.8)

A.12 Demanding housing target with high housing bonus

$I = 25$ KSEK/month, $P = 2.5$ MSEK, $b = 0.7$

Run	Beta (B,H,K,L)	Avg first hitting time (t)	Avg mean(B)	Avg mean(H)	Avg mean(K)	Avg mean(L)	Avg mean cumulative spending	MC success rate (%)	First hitting time distribution	Avg policy shares
1	(0.30,0.40,0.10,0.20)	6.12	59069	665080	185524	380412	322434	99.5	(0,0,0,0,0,2628,559,656,896,181,57)	(3.4,43.8,32.9,19.9)
2	(0.25,0.40,0.15,0.20)	7.26	60370	605549	202818	348750	392658	99.1	(0,0,0,0,0,158,197,3215,992,350,44)	(3.3,40.1,38.6,18.0)
3	(0.20,0.50,0.15,0.15)	5.66	58028	683619	206445	319289	335022	99.7	(0,0,0,0,0,3016,1351,104,335,162,19)	(3.2,45.1,35.1,16.6)

A.13 Demanding housing target with very high housing bonus

$I = 25$ KSEK/month, $P = 2.5$ MSEK, $b = 0.8$

Run	Beta (B,H,K,L)	Avg first hitting time (t)	Avg mean(B)	Avg mean(H)	Avg mean(K)	Avg mean(L)	Avg mean cumulative spending	MC success rate (%)	First hitting time distribution	Avg policy shares
1	(0.30,0.40,0.10,0.20)	6.12	60973	656189	188846	367720	338158	99.8	(0,0,0,0,0,1529,2195,588,557,78,44)	(3.5,43.3,34.0,19.2)
2	(0.25,0.40,0.15,0.20)	5.85	59092	668755	199144	337219	340728	99.4	(0,0,0,0,0,3105,406,827,395,206,31)	(3.2,44.2,35.0,17.6)
3	(0.20,0.50,0.15,0.15)	5.59	57536	687297	206879	317035	333355	99.8	(0,0,0,0,0,3133,1328,118,290,113,9)	(3.1,45.4,35.1,16.5)

A.14 Demanding housing target without housing bonus

$I = 25$ KSEK/month, $P = 2.5$ MSEK, $b = 0$

Run	Beta (B,H,K,L)	Avg first hitting time (t)	Avg mean(B)	Avg mean(H)	Avg mean(K)	Avg mean(L)	Avg mean cumulative spending	MC success rate (%)	First hitting time distribution	Avg policy shares
1	(0.30,0.40,0.10,0.20)	8.65	58625	563261	187643	392651	414915	98.2	(0,0,0,0,0,0,0,2325,2002,585)	(3.2,37.0,39.4,20.4)
2	(0.25,0.40,0.15,0.20)	8.54	56709	563277	199536	369409	427495	97.5	(0,0,0,0,0,0,0,2703,1731,440)	(3.1,36.9,40.9,19.1)
3	(0.20,0.50,0.15,0.15)	8.31	58503	575084	210165	336776	428688	99.3	(0,0,0,0,0,0,0,203,3211,1373,179)	(3.2,37.9,41.6,17.3)

A.15 Constrained income scenario with reduced housing bonus

$I = 17$ KSEK/month, $P = 1.5$ MSEK, $b = 0.4$

Run	Beta (B,H,K,L)	Avg first hitting time (t)	Avg mean(B)	Avg mean(H)	Avg mean(K)	Avg mean(L)	Avg mean cumulative spending	MC success rate (%)	First hitting time distribution	Avg policy shares
1	(0.30,0.40,0.10,0.20)	8.13	41631	298449	95194	188467	176035	23.5	(0,0,0,0,0,0,444,135,596,1)	(4.1,40.2,35.9,19.9)
2	(0.25,0.40,0.15,0.20)	8.99	38138	291597	98897	187536	184983	14.6	(0,0,0,0,0,0,5,724,0)	(3.5,38.9,37.8,19.8)
3	(0.20,0.50,0.15,0.15)	8.73	37980	293038	104751	181018	183896	18.3	(0,0,0,0,0,0,11,228,678,0)	(3.3,39.5,38.3,18.9)

A.16 Constrained income scenario with increased housing bonus

$I = 17$ KSEK/month, $P = 1.5$ MSEK, $b = 0.6$

Run	Beta (B,H,K,L)	Avg first hitting time (t)	Avg mean(B)	Avg mean(H)	Avg mean(K)	Avg mean(L)	Avg mean cumulative spending	MC success rate (%)	First hitting time distribution	Avg policy shares
1	(0.30,0.40,0.10,0.20)	7.16	41246	307571	99866	194085	157647	92.6	(0,0,0,0,0,0,4048,402,175,3)	(4.0,41.6,33.7,20.7)
2	(0.25,0.40,0.15,0.20)	8.65	38399	299151	98322	185642	179591	22.1	(0,0,0,0,0,0,67,257,780,2)	(3.5,40.4,36.8,19.3)
3	(0.20,0.50,0.15,0.15)	7.11	38061	305853	110522	178389	166169	86.2	(0,0,0,0,0,0,172,3681,279,177,1)	(3.4,41.5,36.3,18.8)

A.17 Higher income and demanding housing target with increased housing bonus

$I = 35$ KSEK/month, $P = 2.5$ MSEK, $b = 0.6$

Run	Beta (B,H,K,L)	Avg first hitting time (t)	Avg mean(B)	Avg mean(H)	Avg mean(K)	Avg mean(L)	Avg mean cumulative spending	MC success rate (%)	First hitting time distribution	Avg policy shares
1	(0.30,0.40,0.10,0.20)	3.02	78442	413342	276403	1207820	640205	100.0	(0,0,0,4916,70,14,0,0,0,0,0)	(3.0,17.5,38.7,40.7)
2	(0.25,0.40,0.15,0.20)	3.03	84993	411323	299378	1039854	749800	100.0	(0,0,0,4852,131,17,0,0,0,0,0)	(3.3,17.5,44.2,35.0)
3	(0.20,0.50,0.15,0.15)	3.04	85401	415031	341062	949180	786173	100.0	(0,0,0,4800,180,20,0,0,0,0,0)	(3.3,17.6,47.4,31.7)

A.18 Higher income and demanding housing target with reduced housing bonus

$I = 35$ KSEK/month, $P = 2.5$ MSEK, $b = 0.4$

Run	Beta (B,H,K,L)	Avg first hitting time (t)	Avg mean(B)	Avg mean(H)	Avg mean(K)	Avg mean(L)	Avg mean cumulative spending	MC success rate (%)	First hitting time distribution	Avg policy shares
1	(0.30,0.40,0.10,0.20)	3.02	78442	413342	276403	1207820	640205	100.0	(0,0,0,4916,70,14,0,0,0,0,0)	(3.0,17.5,38.7,40.7)
2	(0.25,0.40,0.15,0.20)	3.08	84909	413944	299267	1036851	749248	100.0	(0,0,0,4628,355,17,0,0,0,0,0)	(3.3,17.6,44.2,34.9)
3	(0.20,0.50,0.15,0.15)	3.08	85327	417014	341306	942997	788724	100.0	(0,0,0,4632,348,20,0,0,0,0,0)	(3.3,17.7,47.5,31.5)

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